Earthquake Deformation - Turkey 2023

Sentinel-1 SAR Interferometry Analysis

Project Report

Submitted as part of the coursework for

Advanced Remote Sensing

Assignment #2

at the Faculty of Digital and Analytical Sciences

of the Paris-Lodron-University of Salzburg

submitted by

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1. Introduction

The goal of the second assignment was to create and interpret interferometric products over the same area of interest. This specific example used Sentinel-1 SAR datasets. A differential interferogram and displacement map were created in the first part of the assignment, to highlight surface deformation that was related to the chosen earthquake. For this interferogram, it was necessary to choose images before and after the hazard. After that, a topographic interferogram and DEM were created to represent topographic phase before the earthquake occurred. Both images that were used in topographic interferogram had to date before the hazard.

2. Study Area

Initially, the author chose earthquake that happened in Zagreb in 2020, because the building from his previous Geodesy and Geoinfomatics faculty got damaged. The results were not significant, so the study area changed to the earthquake that occurred in Turkey, close to the Syrian border. The earthquake happened at around 4AM on 6th February 2023. The epicenter with a registered magnitude of 7.8 was located 37km west-northwest from the Gaziantep city in Turkey (close to Syrian border). The impacts were severe as over 50,000 people lost their life and over 100,000 people were injured. Thousands of buildings got destroyed, as well as roads damaged. But the hazard did not stop there, as the earthquake was followed with afterschocks (Internet Geography). Surface deformations caused by the earthquake were investigated using created differential interferogram and displacement map, and the terrain characteristics of AOI were investigated using topographic interferogram and DEM.



Figure 1. Collapsed buildings in Hatay (Turkey) as a result of the earthquake (NPR)



Figure 2. Local man praying in front of the ruins (The Economist)

3. Datasets Download

The epicenter coordinates were different across the sources, and the coordinates provided by GeoHack (Wikimedia Cloud Services) were taken into consideration.

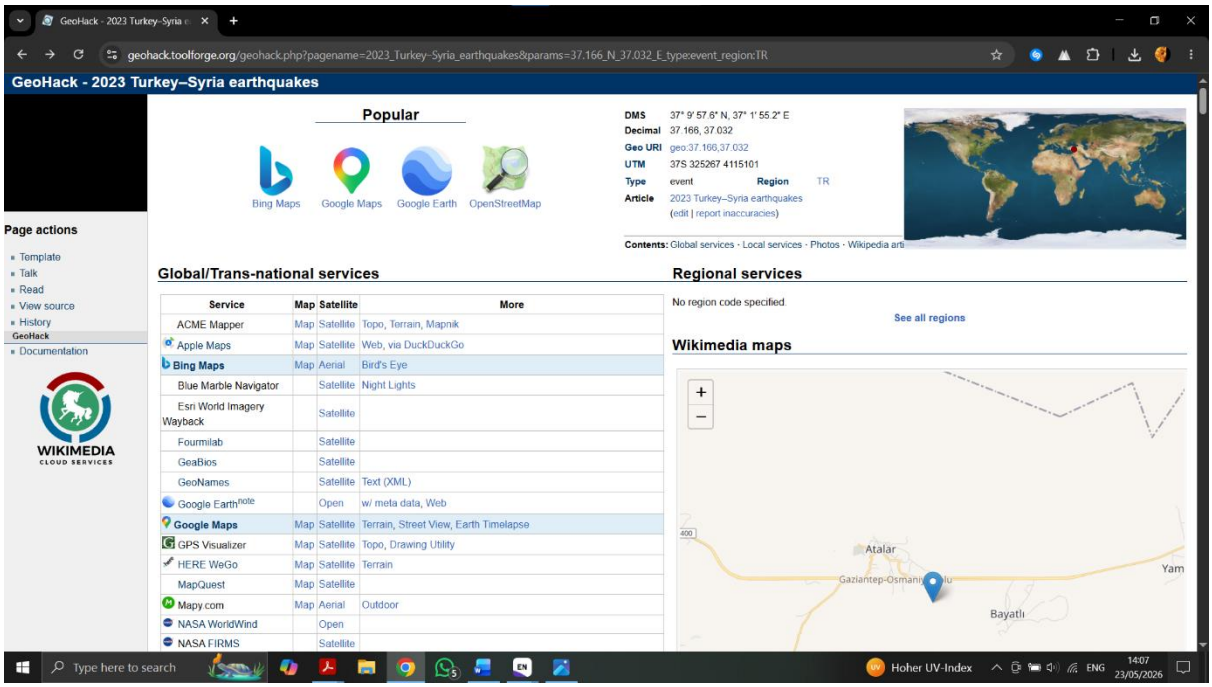


Figure 3. Searching for the earthquake epicenter coordinates on GeoHack (Wikimedia Cloud Services)

Later, the satellite images for the interferometric analysis were downloaded from the ASF Data Search platform (Alaska Satellite Facility). The area around the epicenter was defined using the rectangle selection option. In the filter section, the staring date was defined as 9th January 2023, and the ending date was defined as 9th February 2023. For the file type was selected as L1 Single Look Complex (SLC), the beam mode

was selected as IW. After the appropriate dataset was found (9th February 2023), “More like this” was selected so the datasets with similar/same metadata are shown. Three datasets were downloaded, where the images acquired on 28th January 2023 and 9th February 2023 were used to generate a differential interferogram, and the images acquired on 16th January 2023 and 28th January 2023 were used for topographic interferogram.

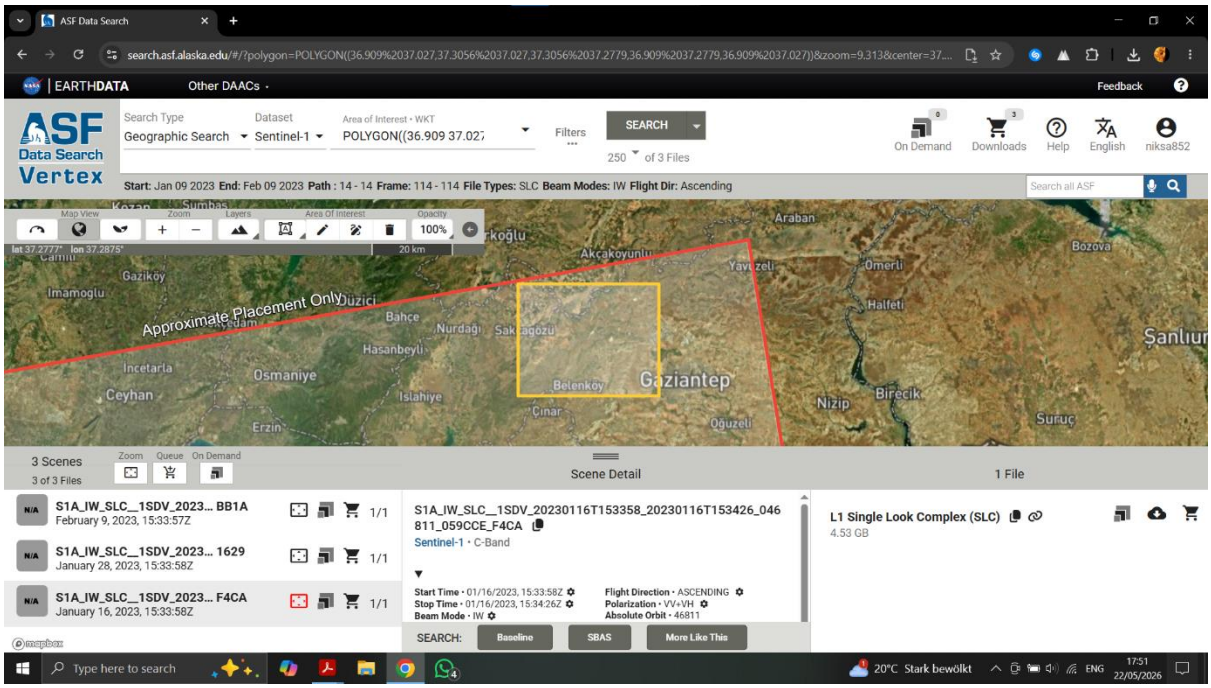


Figure 4. Selecting the appropriate datasets in the ASF Data Search platform (Alaska Satellite Facility)

4. Differential Interfegoram & Displacement Map

For the technical part, SNAP open source sofware was used which is developed by European Space Agency (ESA). The differential interferogram and displacement map were created to investigate possible surface displacement. Satellite image acquired on 28th January was used as the dataset before the earthquake, and the satellite image acquired on 9th February was used as the dataset after the earthquake.

a) Loading Sentinel-1 images (Data Preparation)

All three datasets were imported into SNAP using the “Open Product” option (two images that were used for differential interferogram + one image that was later used for topographic interferogram).

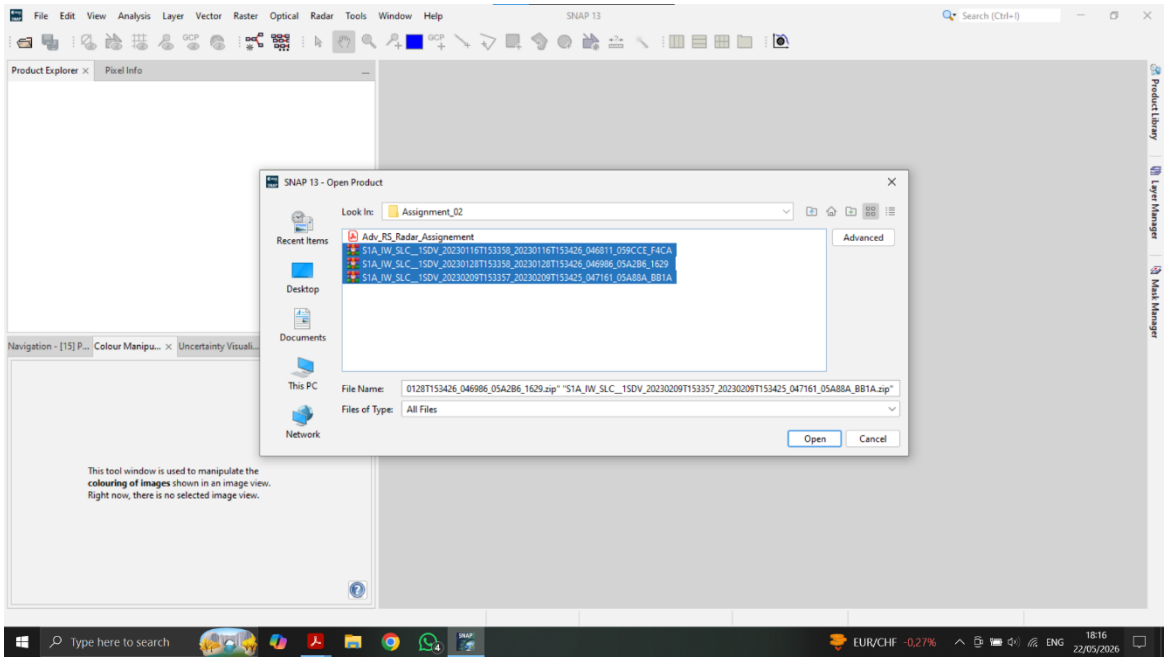


Figure 5. Importing all three downloaded datasets into SNAP (Author)

b) Splitting the Subswath (Data Preparation)

After that, it was necessary to pre-process the datasets that will be used for creating the differential interferogram. As a first step, it was required to choose the subswath and burts that include the area of interest. It was selected “Radar”, “Sentinel-1 TOPS” and then “S-1 TOPS Split”. IW3 was chosen for the subswath, polarisation method was selected as VV, and the bursts were selected from 6 to 9. The target product names were shortened so they do not get too long by the end of process.

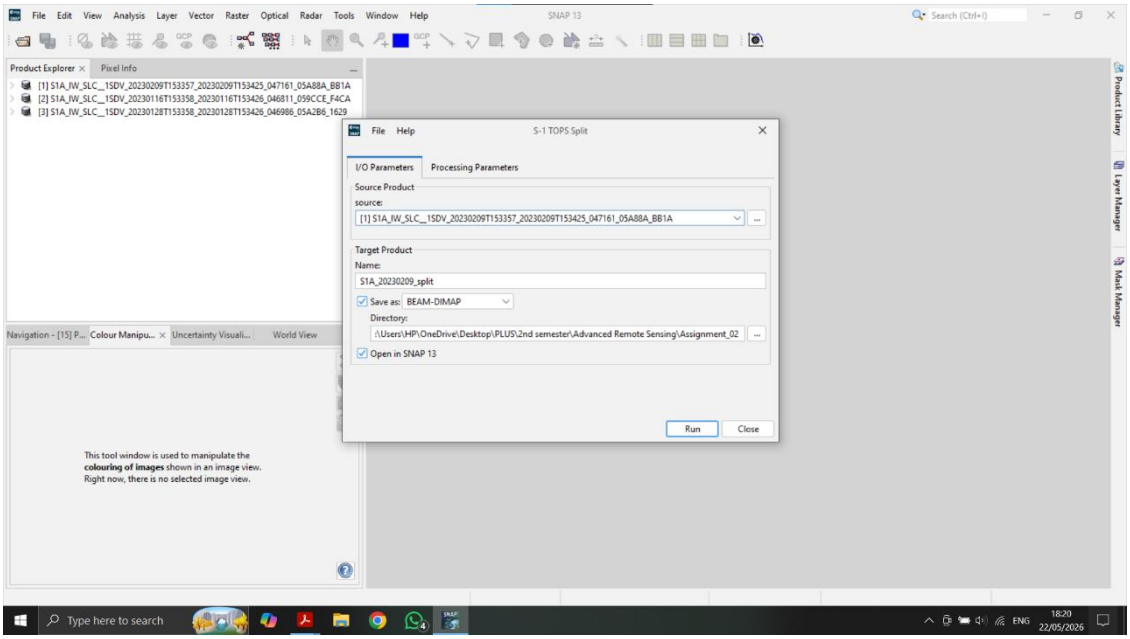


Figure 6. Chosing shorter name for the 9th February dataset while performing "S-1 TOPS Split" (Author)

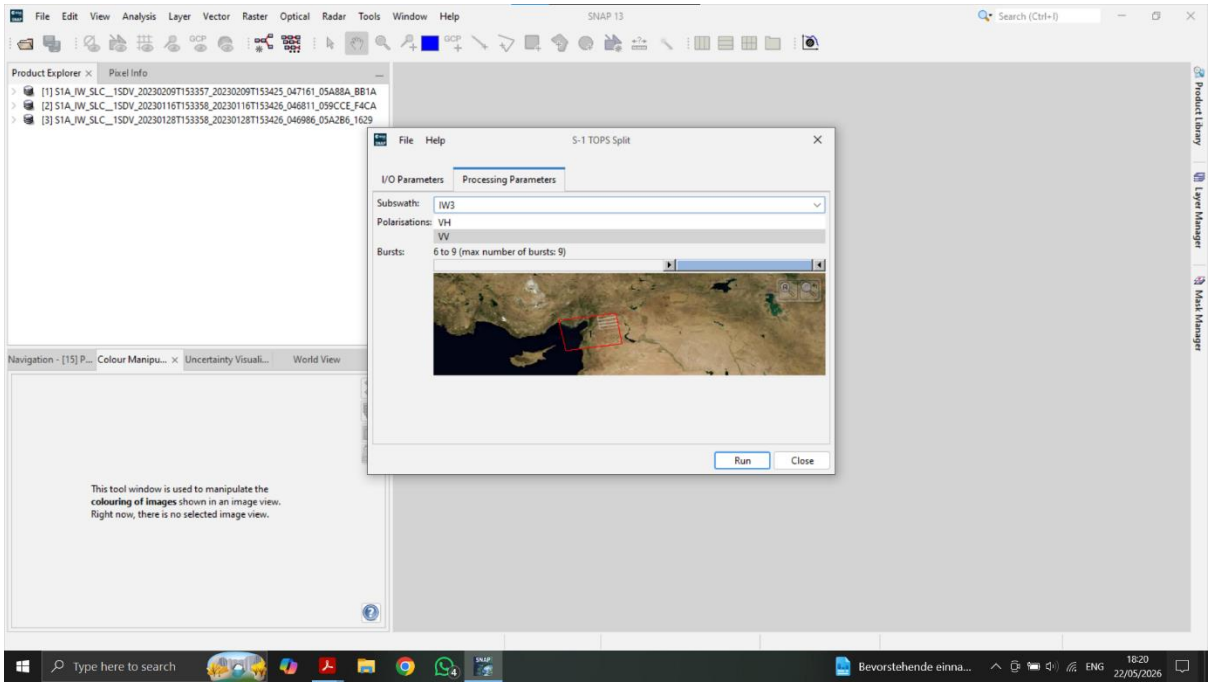


Figure 7. Selecting the subswath, polarisation method and burts that contain AOI for 9th February dataset (Author)

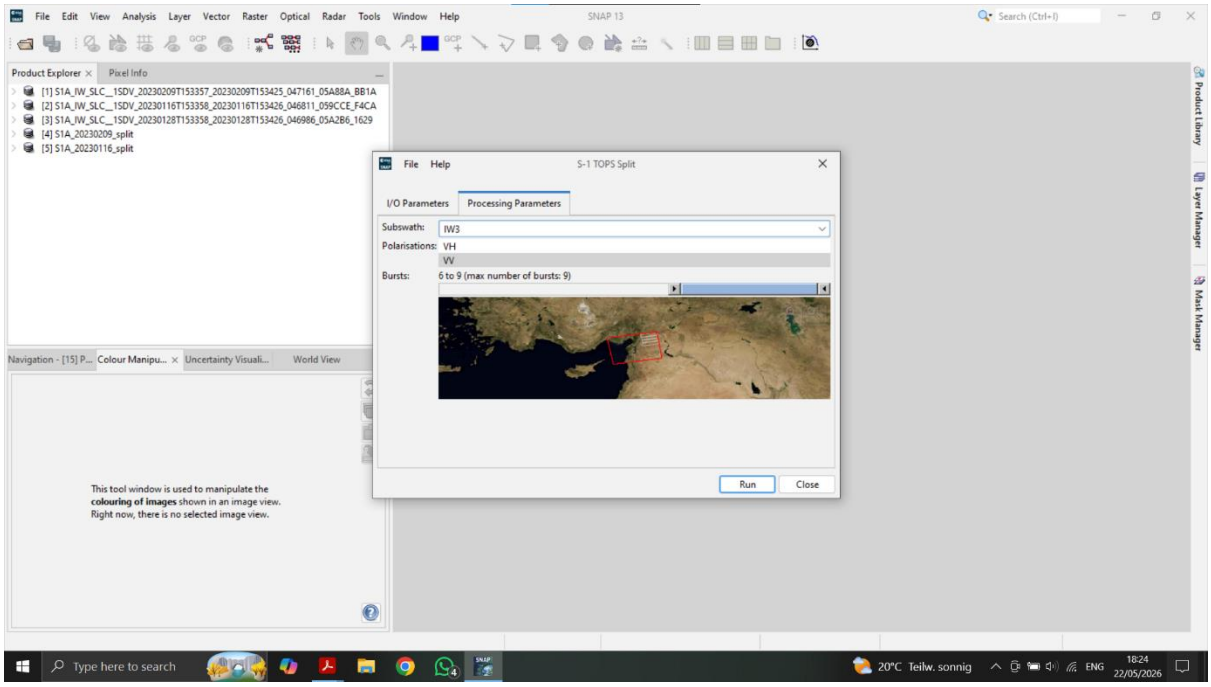


Figure 8. Selecting the subswath, polarisation method and burts that contain AOI for 28th January dataset (Author)

c) Applying Orbit File (Data Preparation)

The last step of data prepration was to correct the satellite positioning errors by selecting “Radar” and then “Apply Orbit File”.

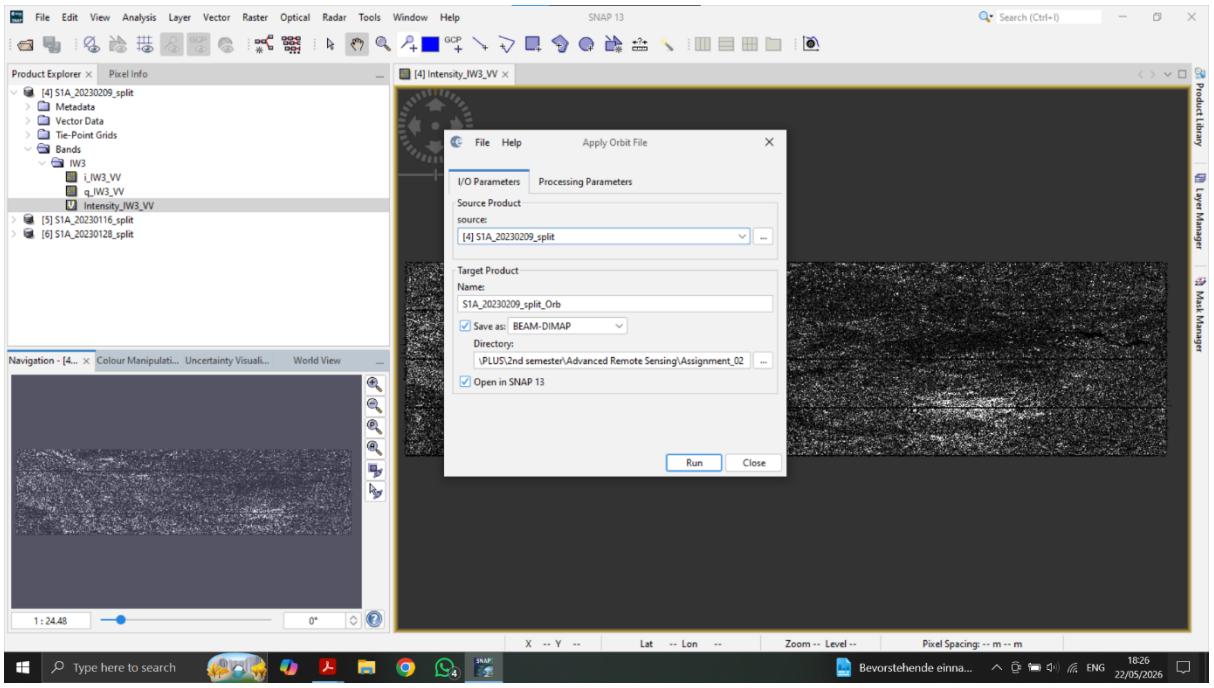


Figure 9. Performing "Apply Orbit File" for 9th February dataset (Author)

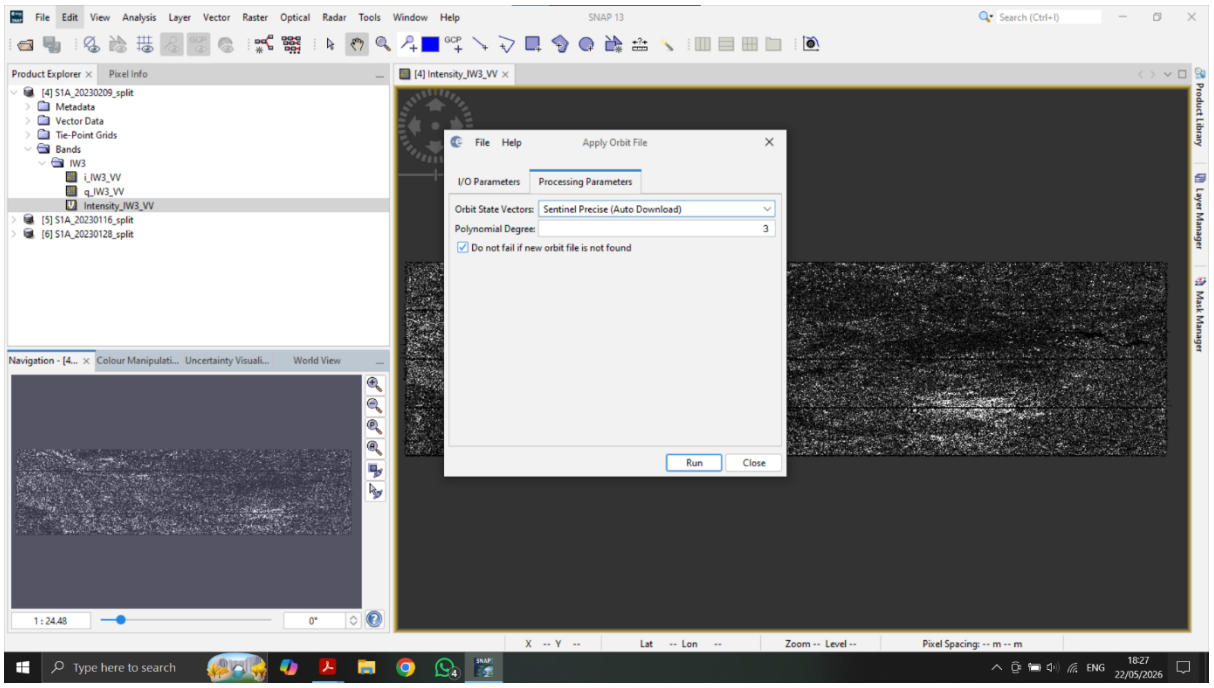


Figure 10. Checking processing parameters in “Apply Orbit File” for 9th February dataset (Author)

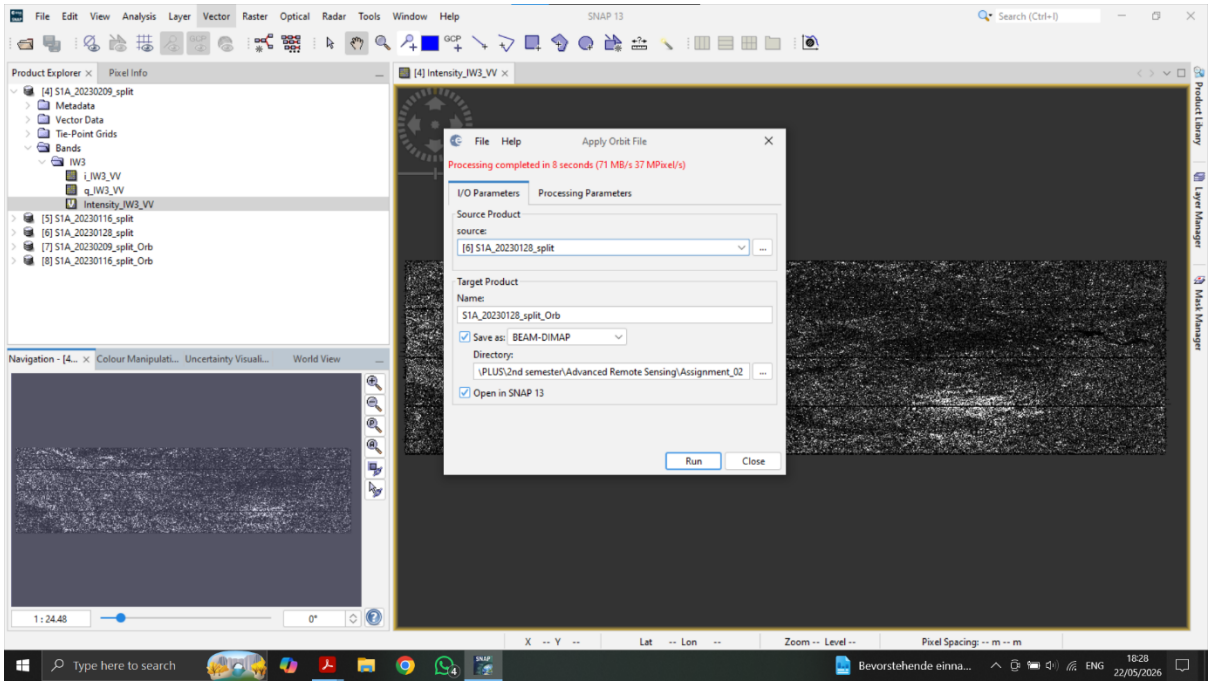


Figure 11. Performing "Apply Orbit File" for 28th January dataset (Author)

d) Back Geocoding (Coregistration)

In this part, it was necessary to align the master and slave images. For this was selected “Radar”, “Coregistration”, “S-1 TOPS Coregistration” and then “S-1 Back Geocoding”. Then, then the master and slave images were added, and DEM was changed to “SRTM 1Sec HGT (Auto Download)”. The rest was kept as it is.

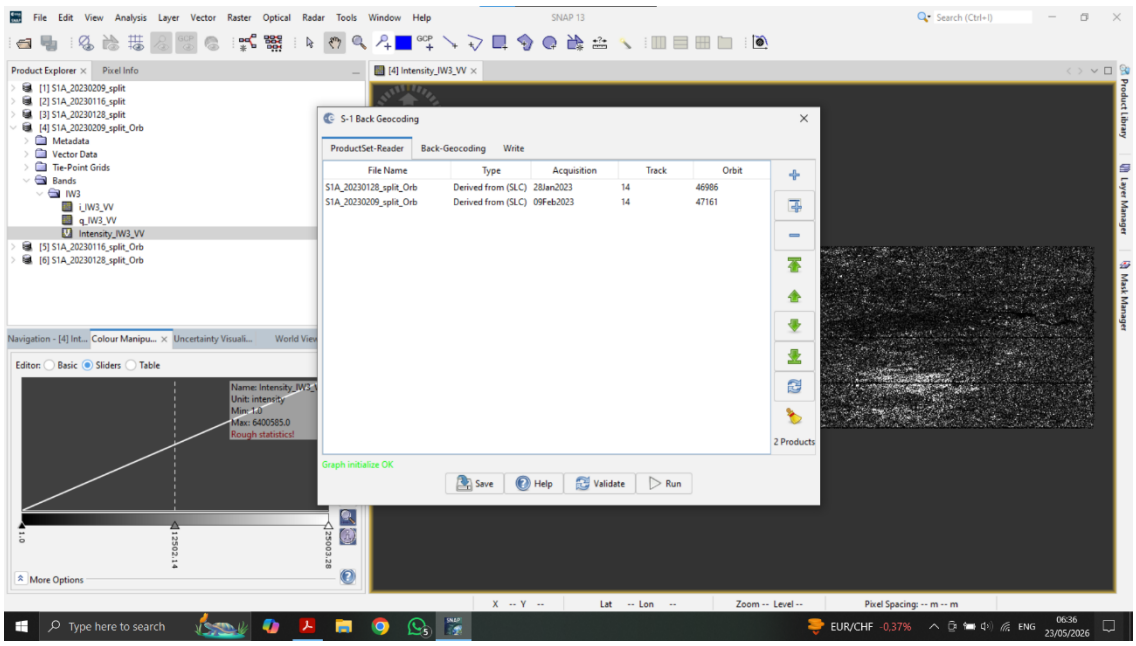


Figure 12. Adding the master and slave images in the "ProductSet-Reader" section (Author)

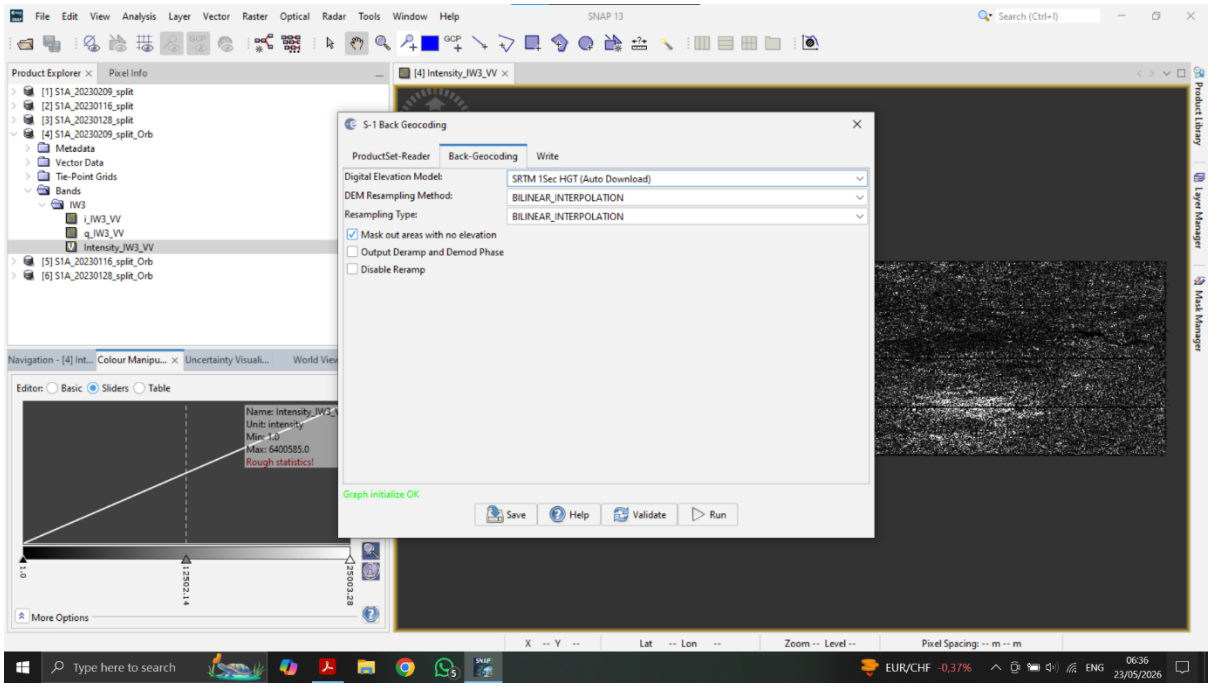


Figure 13. Changing DEM in the "Back-Geocoding" section (Author)

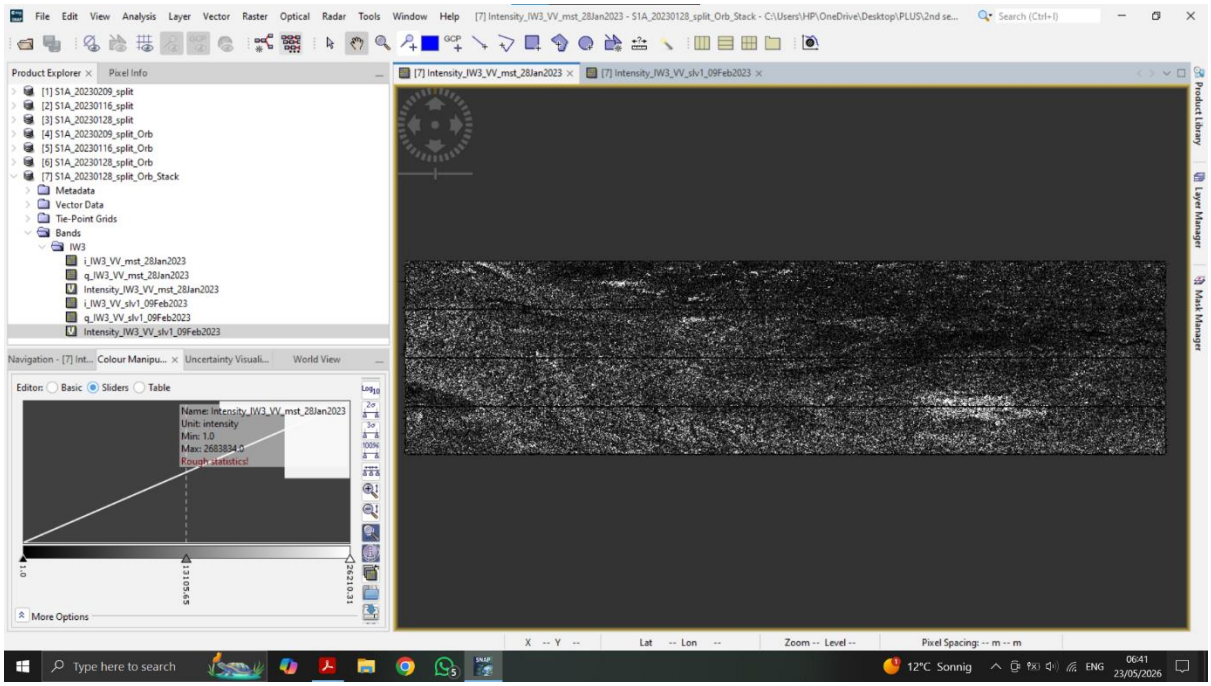


Figure 14. The outcome for 28th January dataset (Author)

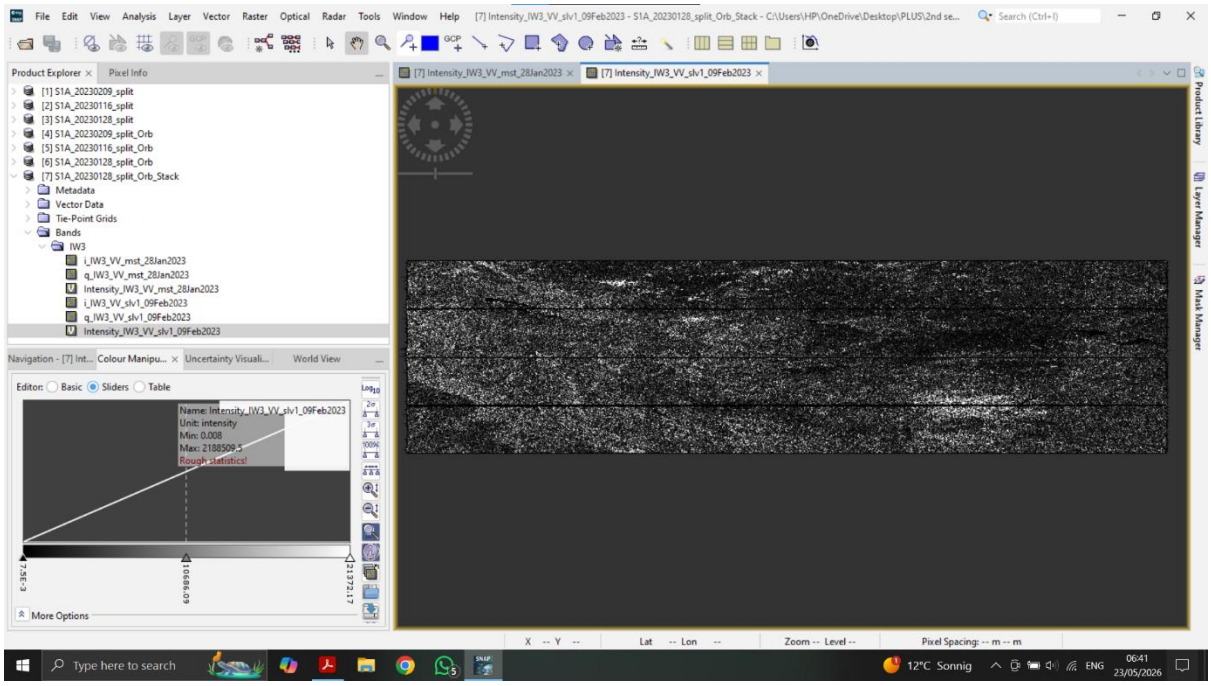


Figure 15. The outcome for 9th February dataset (Author)

e) Enhanced Spectral Diversity (Coregistration)

It was selected “Radar”, “Coregistration” and then “S-1 Enhanced Spectral Diversity”. There was no need for changes in settings.

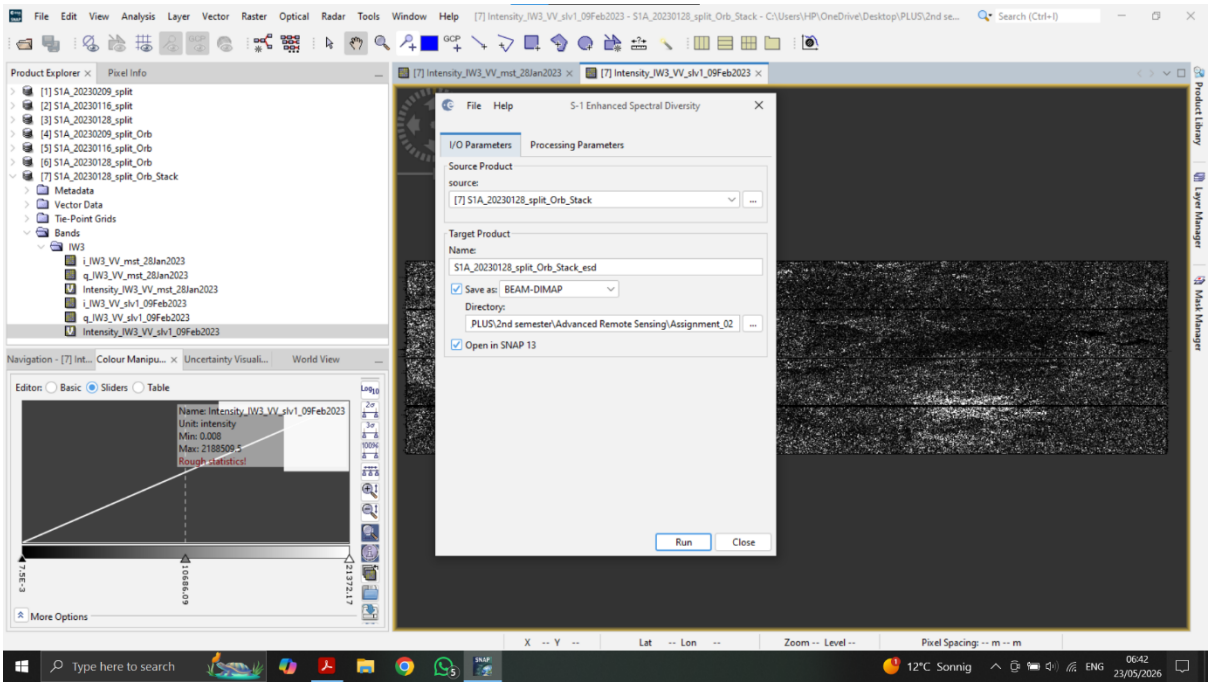


Figure 16. Performing "Enhanced Spectral Diversity” (Author)

f) Interferogram Formation (Topographic Interferogram Generation)

This step included generating interferogram with topographic phase which was later removed. It was selected “Radar”, “Interferometric”, “Products” and then “Interferogram Formation”. For the processing parameters, “Coherence Range Window Size” was changed to 20, and “Coherence Azimuth Window Size” automatically changed to 5. The rest was left as it is.

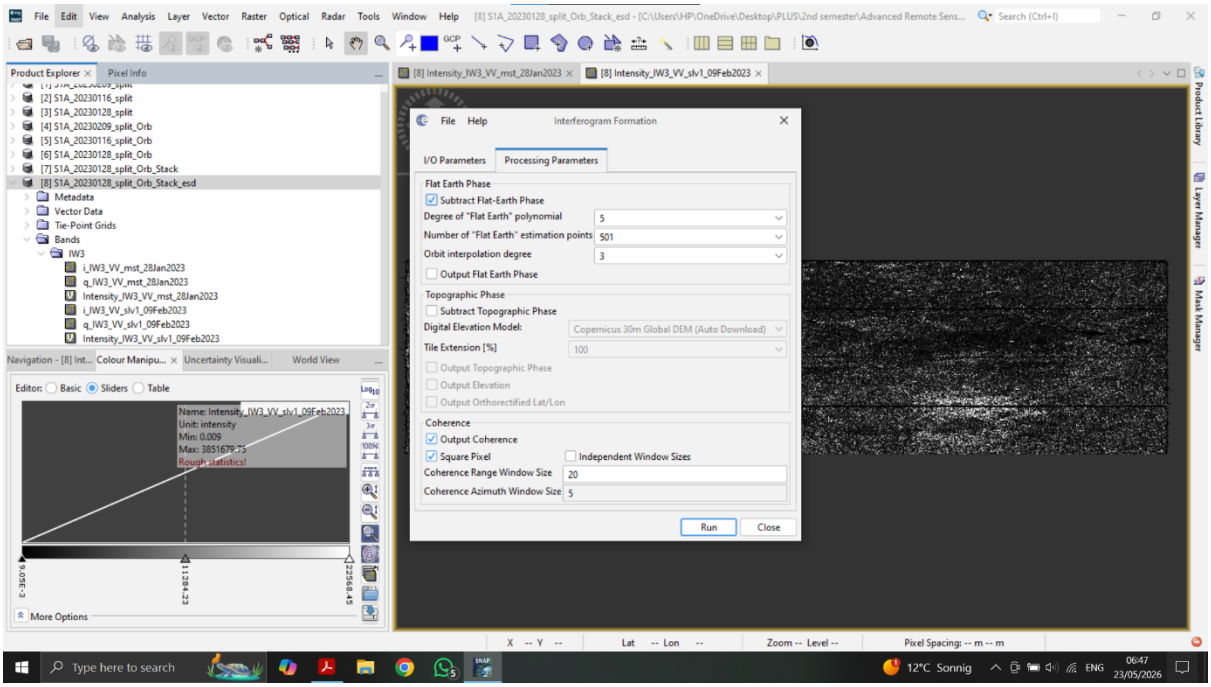


Figure 17. Changing the processing parameters for "Interferogram Formation" (Author)

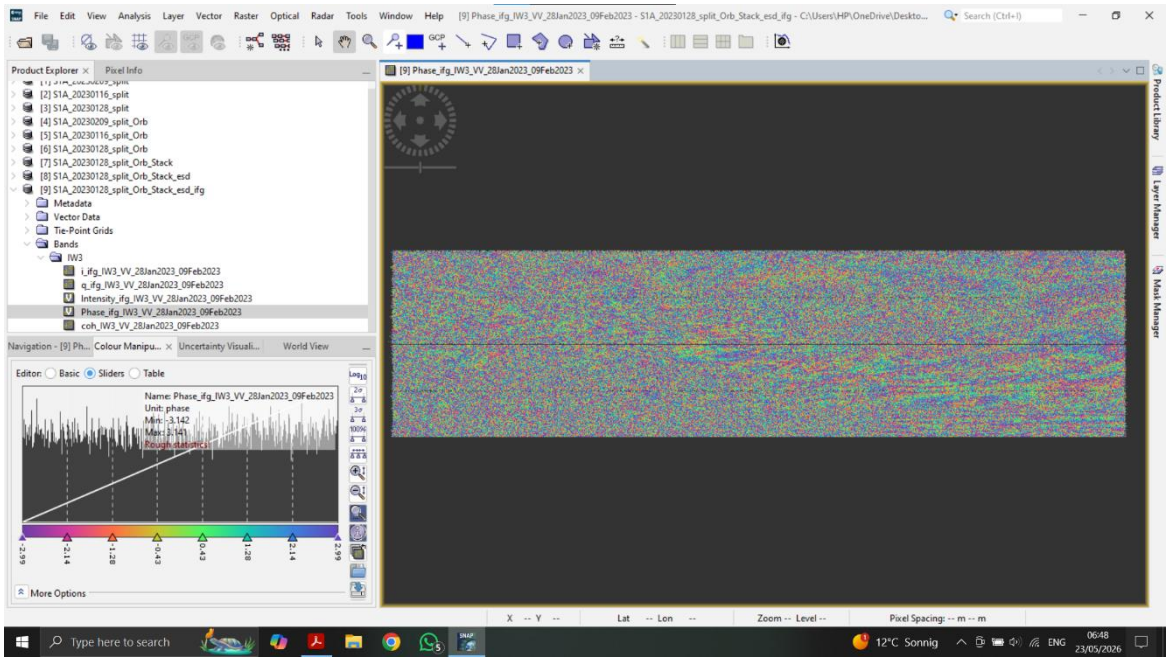


Figure 18. The resulting interferogram (Author)

g) TOPSAR Deburst

Next step was to remove the burst boundaries which resulted in merged interferogram. For this step was selected “Radar”, “Sentinel-1 TOPS” and then “S-1 TOPS Deburst”. The settings were left as they were.

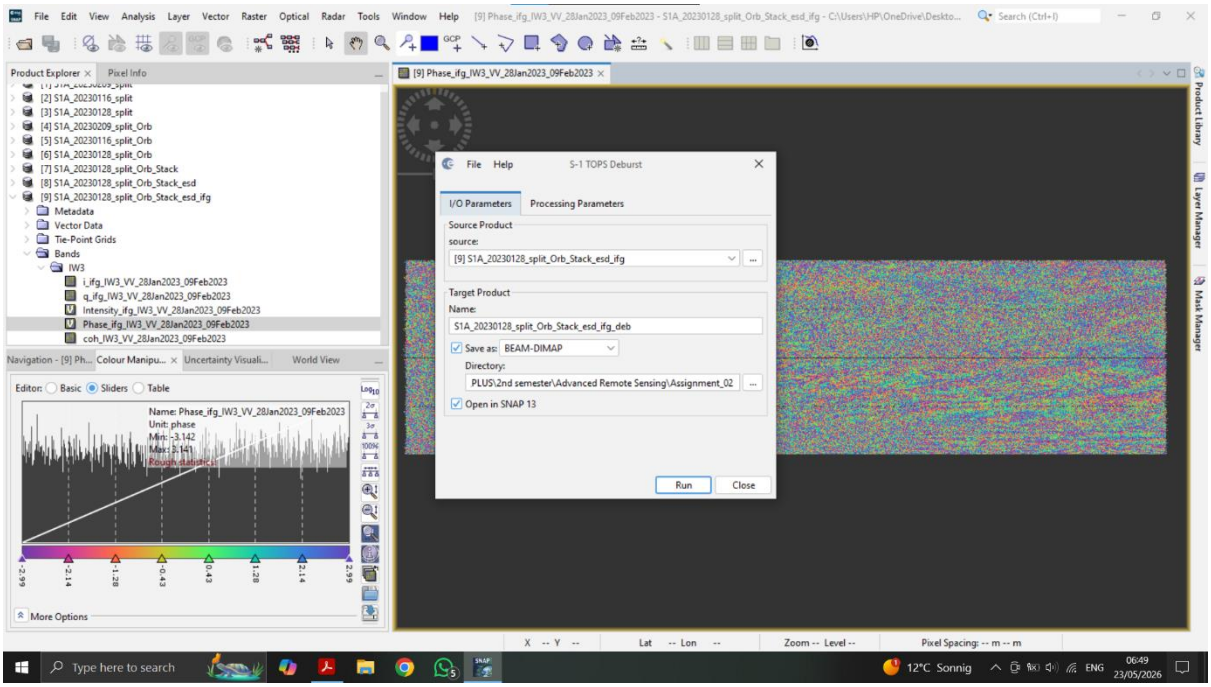


Figure 19. Performing "S-1 TOPS Deburst" (Author)

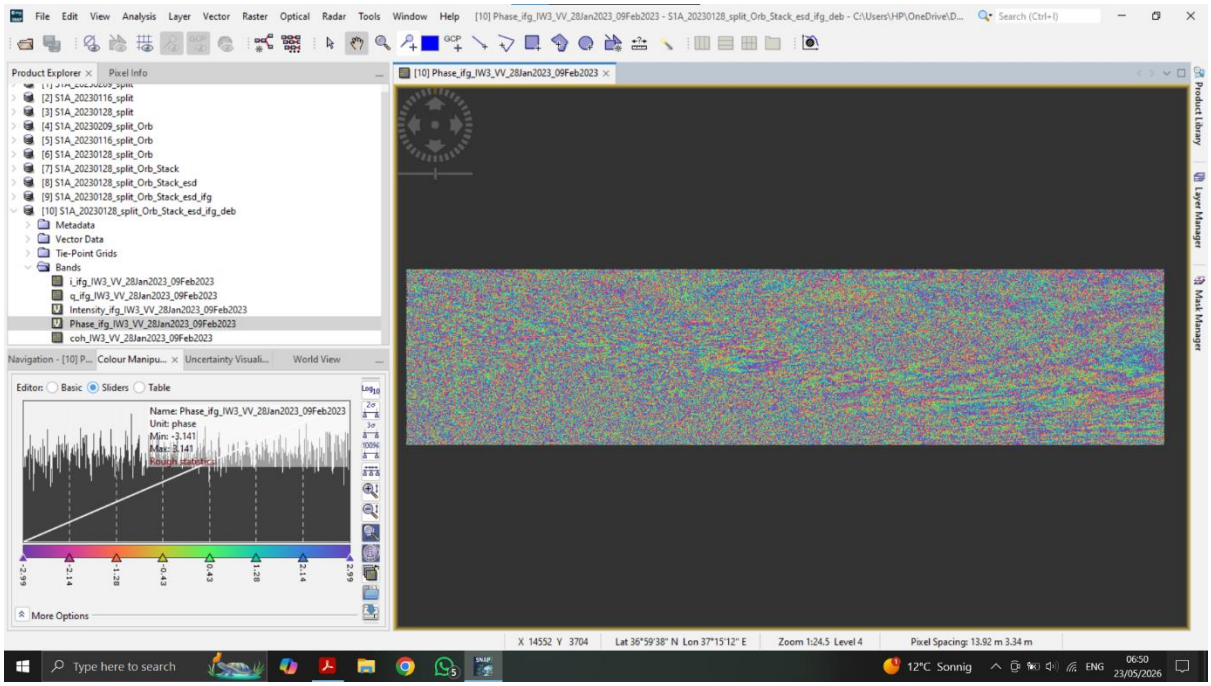


Figure 20. The resulting merged interferogram (Author)

h) Topographic Phase Removal (Differential Interferogram Generation)

This step included removing the topographic effects, so the phase changes caused by ground displacement can be extracted. It was selected “Radar”, “Interferometric”, “Products” and then “Topographic Phase Removal”. In the settings it was required to change DEM to “SRTM 1Sec HGT (Auto Download)” for the topographic correction. For the “Source product” was selected topographic interferogram from the last step.

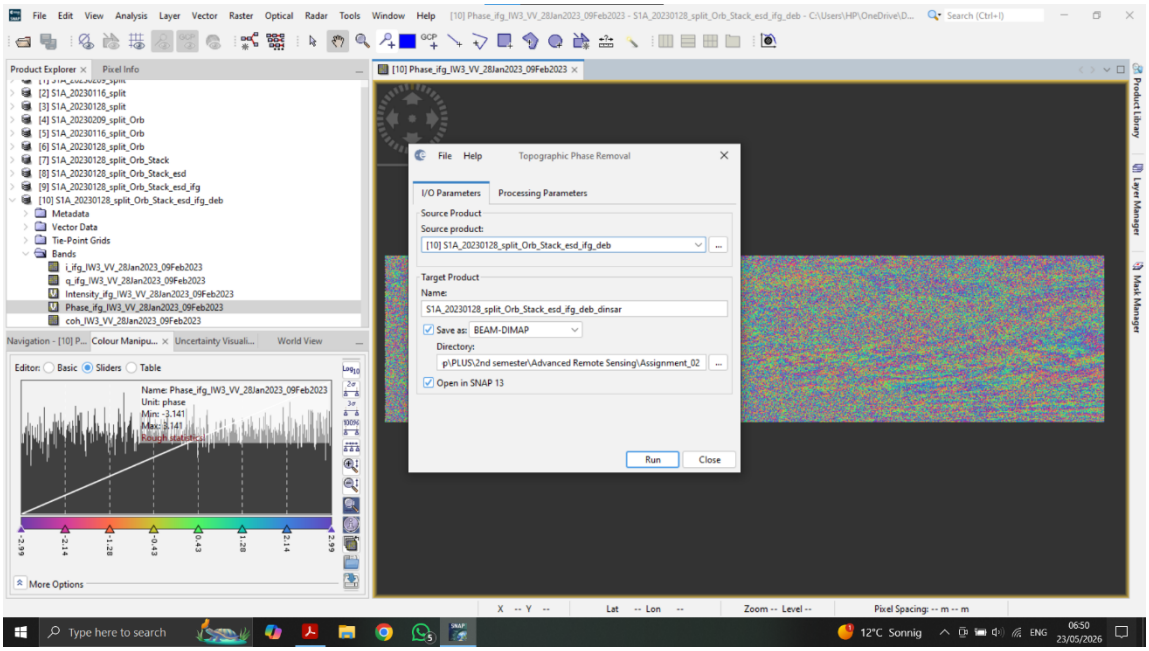


Figure 21. Selecting the topographic interferogram created in the last step as input (Author)

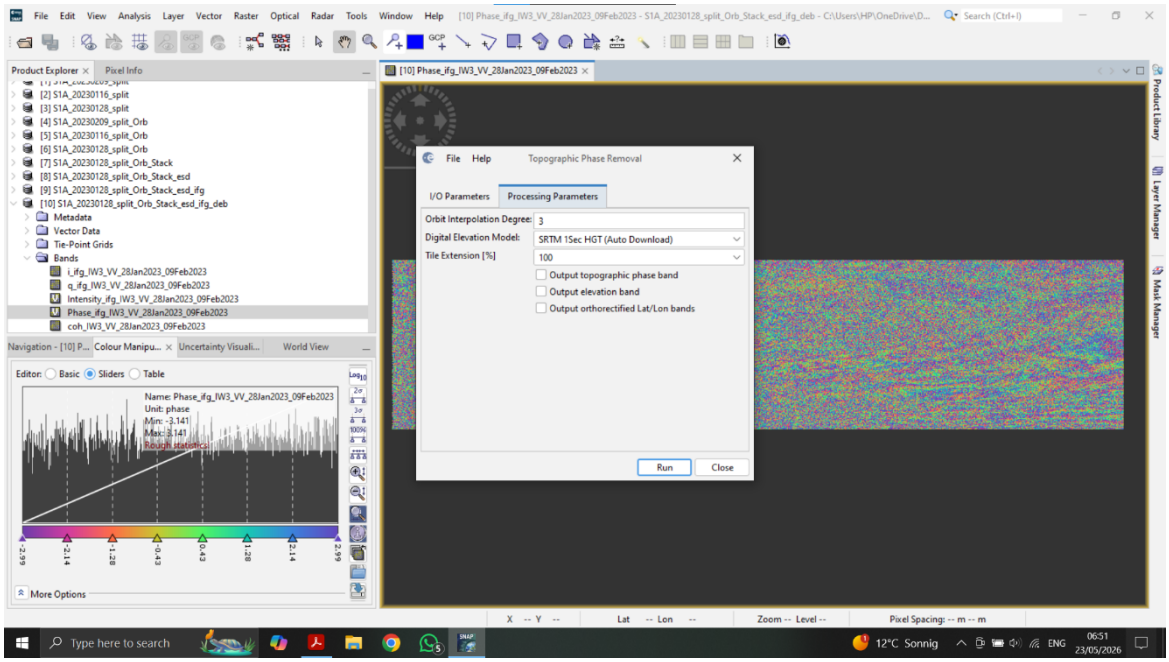


Figure 22. Selecting "SRTM 1Sec HGT (Auto Download)" as DEM (Author)

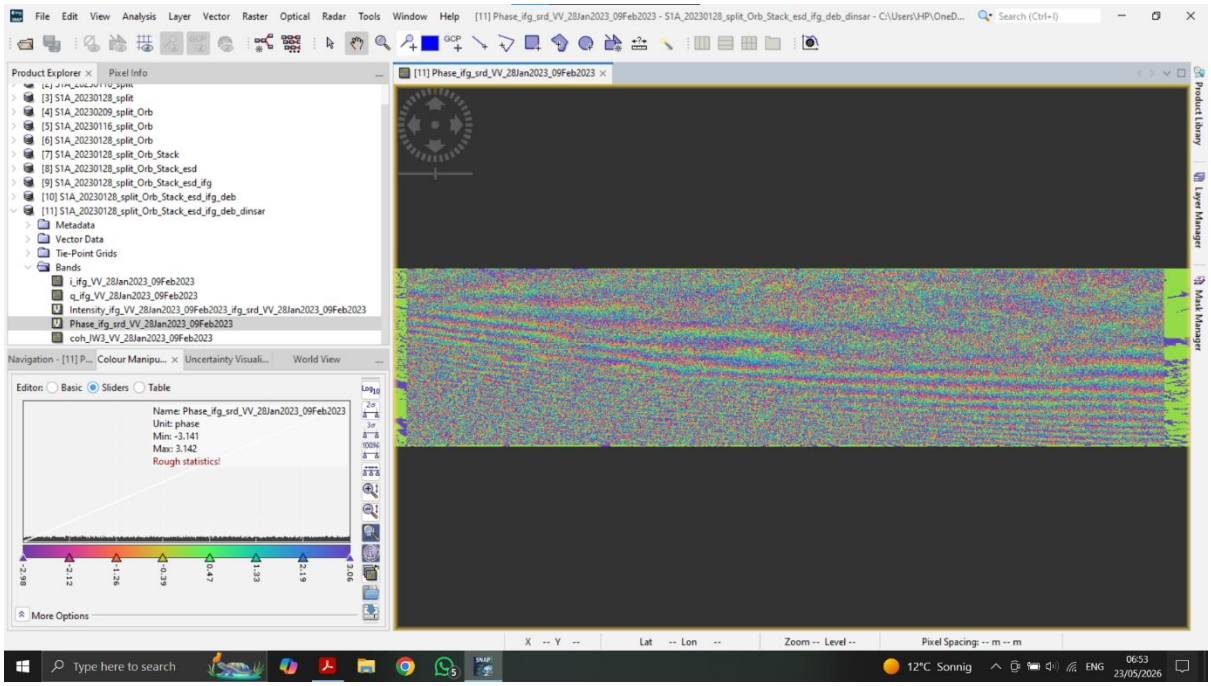


Figure 23. The resulting differential interferogram after removing topographic phase (Author)

i) Multilooking

Next step was to improve visual interpretation of the differential interferogram by performing “Multilooking”, which reduces noise and makes pixels squared. It was selected “Radar”, “SAR Utilities” and then “Multilooking”. In the “Processing Parameters” section, “Number of Range Looks” was changed to 20, and “Number of Azimuth Looks” automatically changed to 5. The input was differential interferogram from the last step. The rest was left as it is.

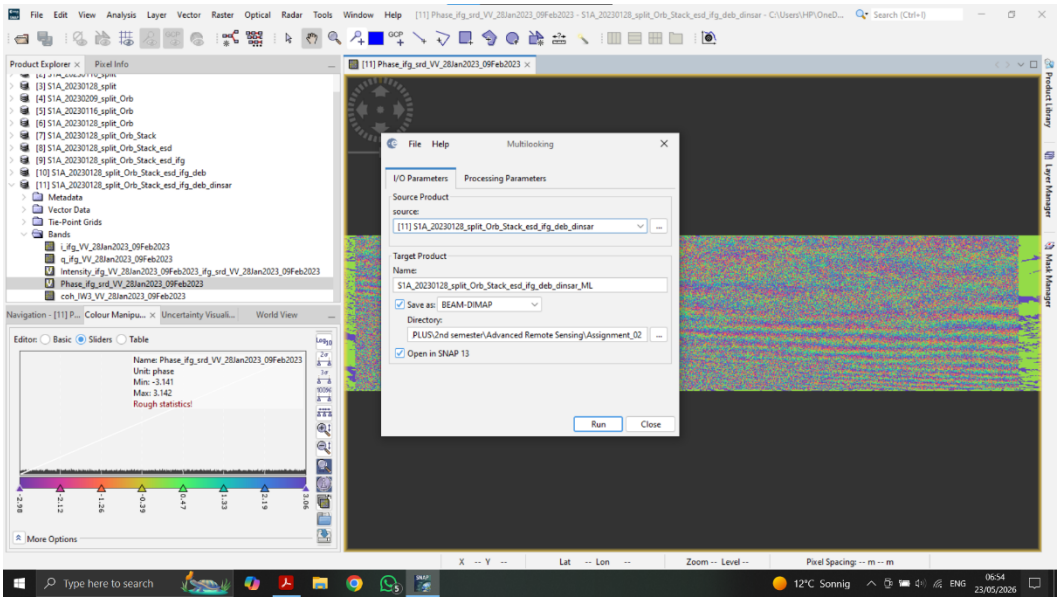


Figure 24. Selecting the differential interferogram from the last step as input (Author)

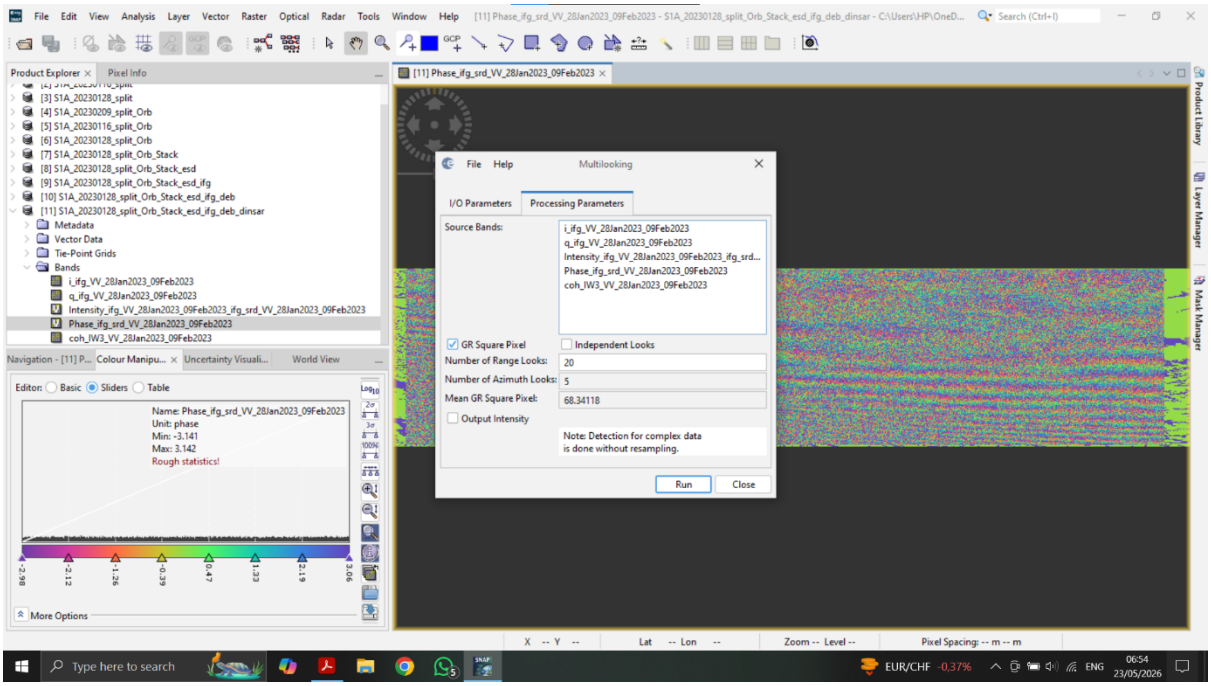


Figure 25. Changing the processing parameters for "Multilooking" (Author)

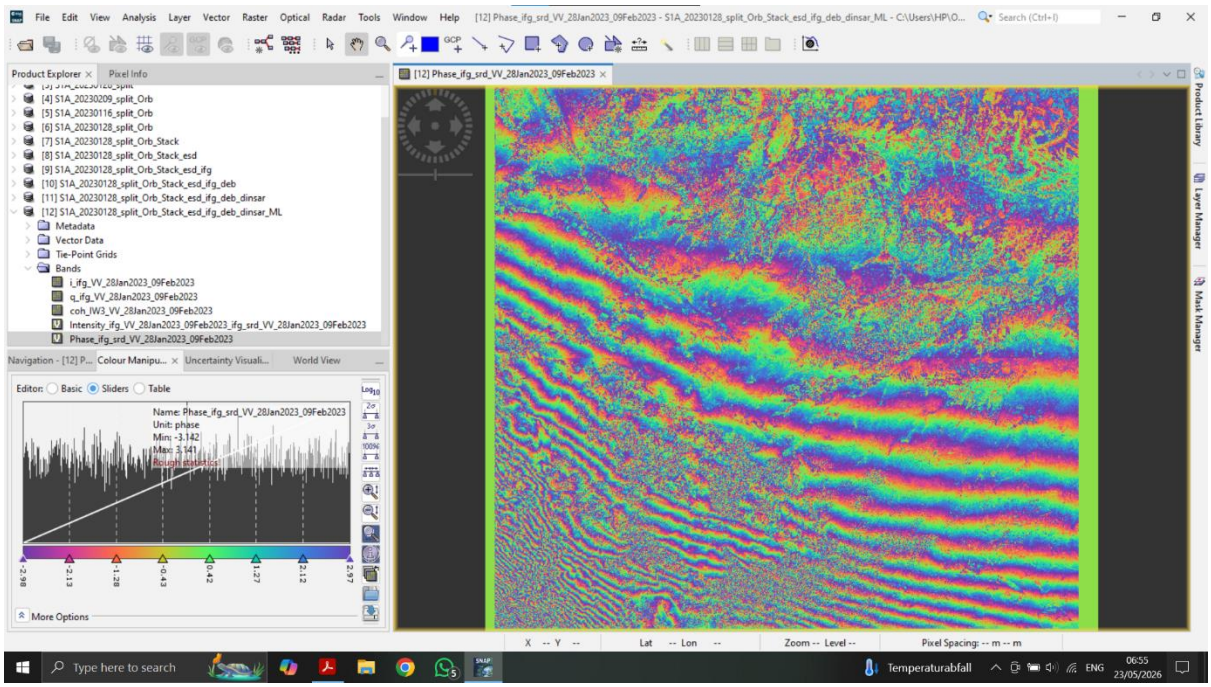


Figure 26. The resulting differential interferogram after performing "Multilooking" (Author)

j) Goldstein Phase Filtering

This was the last step of generating a differential interferogram part, and its purpose was to improve the phase quality (reduce noise) to ensure accurate unwrapping and displacement calculation. “Coherence Threshold” was changed to 0.8 in the processing parameters section. The rest was left as it is.

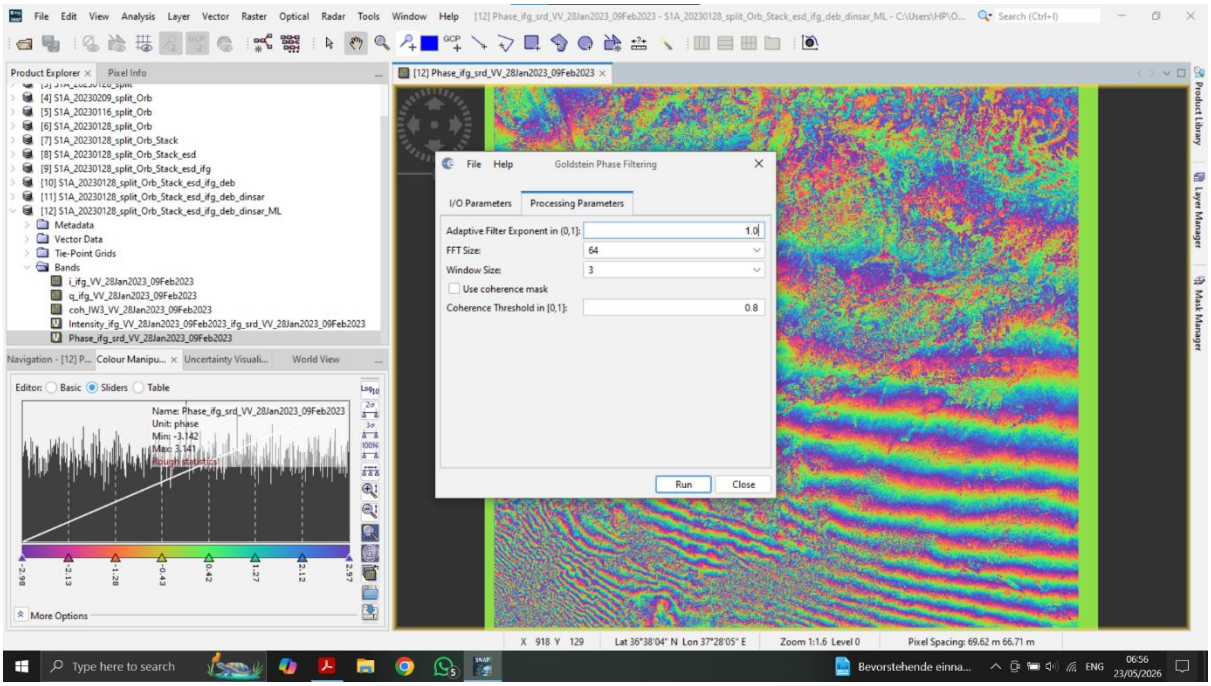


Figure 27. Changing the "Coherence Treshold" value to 0.8 (Author)

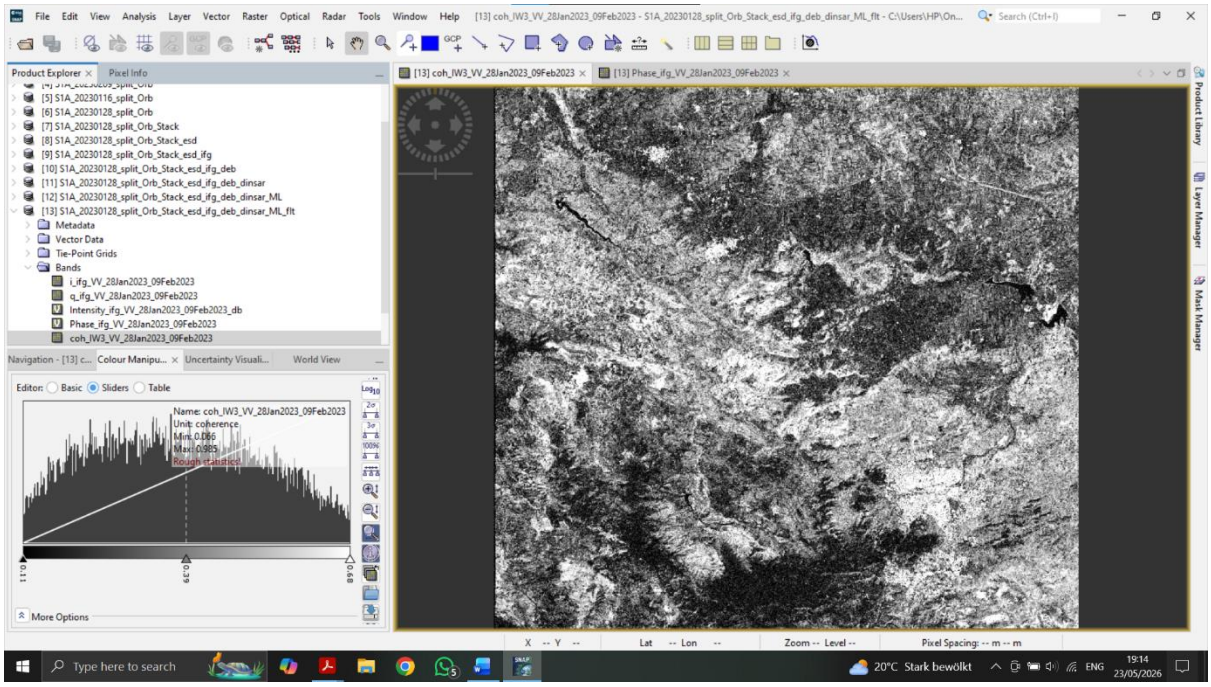


Figure 28. The resulting interferometric coherence after performing “Goldstein Phase Filtering" (Author)

The interferometric coherence measures the quality of the interferometric pair. Areas represented with brighter colors mean that the phase has better quality, and areas represented with darker colors mean that the phase has lower quality. The values range from 0.066-0.985 according to the min and max values in the statistics section. Brighter colors (better quality) can be observed in central and lower right areas. Darker colors (lower quality) can be observed in lower central to lower left and upper right areas.

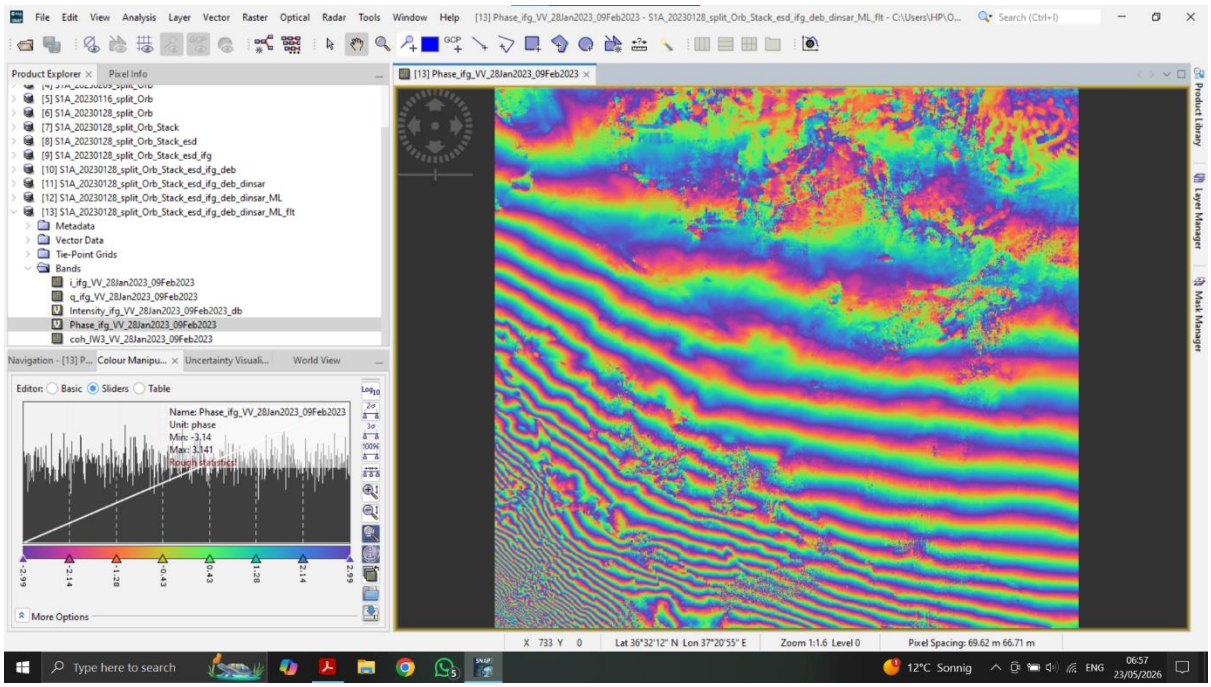


Figure 29. The resulting interferometric phase after performing "Goldstein Phase Filtering" (Author)

As shown in the resulting interferometric phase representation, fringe patterns can be observed in central to lower parts. Their density is the highest in the lower left part and gradually decreases from lower left corner to upper right corner of the representation. The spatial continuity of fringes is more or less solid from central to lower parts. Irregularities can be observed in upper parts. According to both interferometric representations (coherence and phase), the most significant surface displacement across smaller space has occurred in the lower left corner.

k) Snaphu Export

The next step of the analysis was to create a displacement map. In order to perform this, it was necessary to unwrap the phase using Snaphu to obtain continuous displacement measurements. First step was to select “Radar”, “Interferometric”,

“Unwrapping” and then “Snaphu Export”. For the “Source product”, the differential interferogram created in the last step was selected. In the settings, the target folder was defined, “Number of Tile Rows” and “Number of Tile Columns” were set to 1, “Number of Processors” was set to 4, “Row Overlap” and “Column Overlap” were set to 0, and “Tile Cost Threshold” was set to 500. Other settings were not changed.

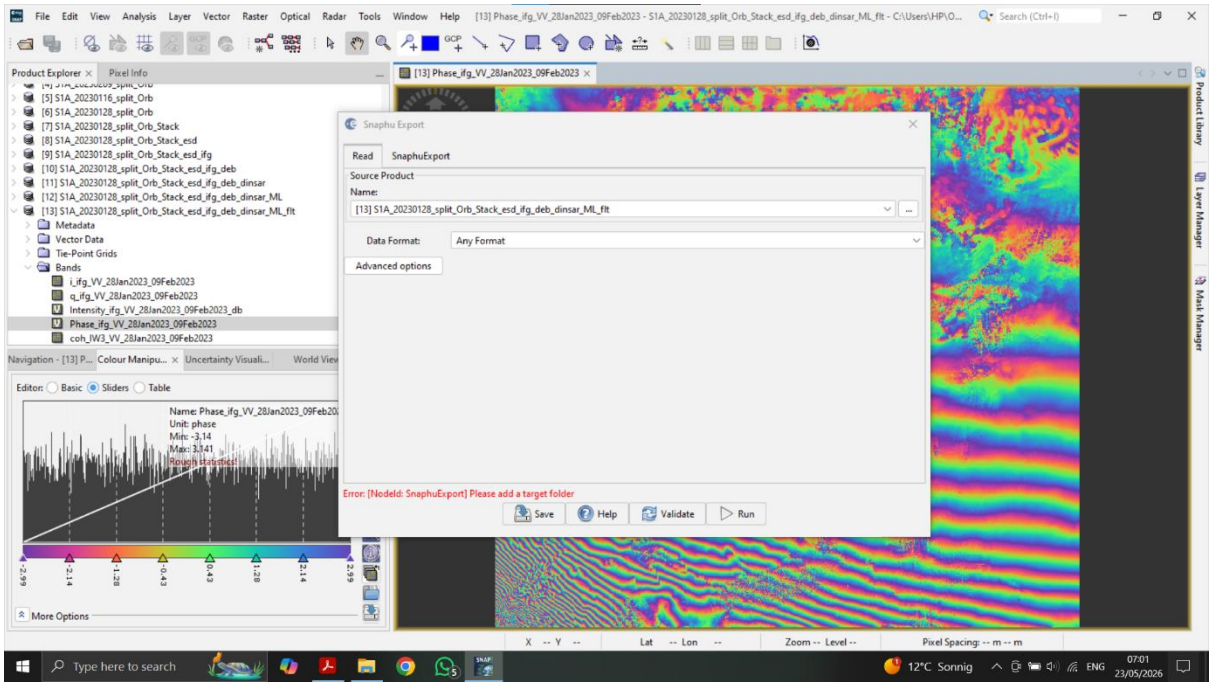


Figure 30. Selecting the differential interferogram from the last step as "Source Product" (Author)

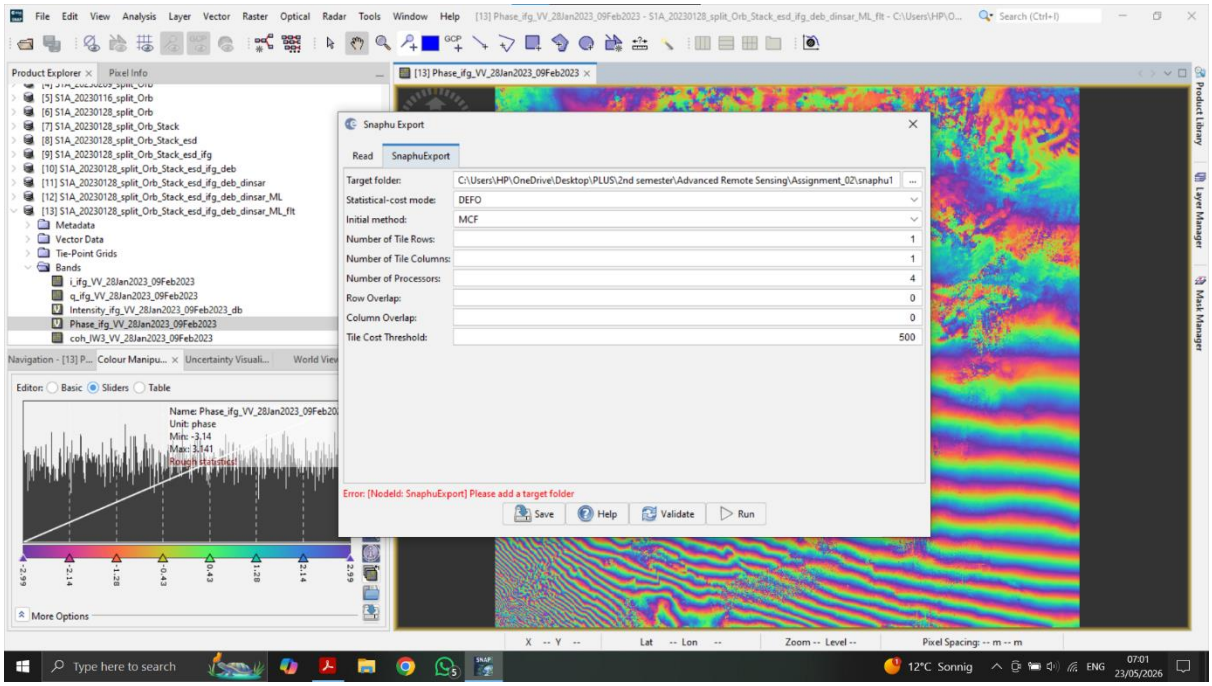


Figure 31. Defining the settings for "Snaphu Export" (Author)

After that, Snaphu was downloaded from the link provided in the task requirements. After extracting the .zip file, the “snaphu” application and “msys-2.0.dll” application extension were placed in the exported “snaphu1” folder. Then, “snaphu” CONF file was opened and “#” was added before “LOGFILE”, and the coherence file name was added to the “CORRFILE” line. Section “snaphu -f snaphu.conf Phase_ifg_VV_28Jan2023_09Feb2023.snaphu.img 1243” was copied and the changes were saved. Next, the copied section was pasted in the command line and executed. When the unwrapping finished, a new “UnwPhase” file occurred.

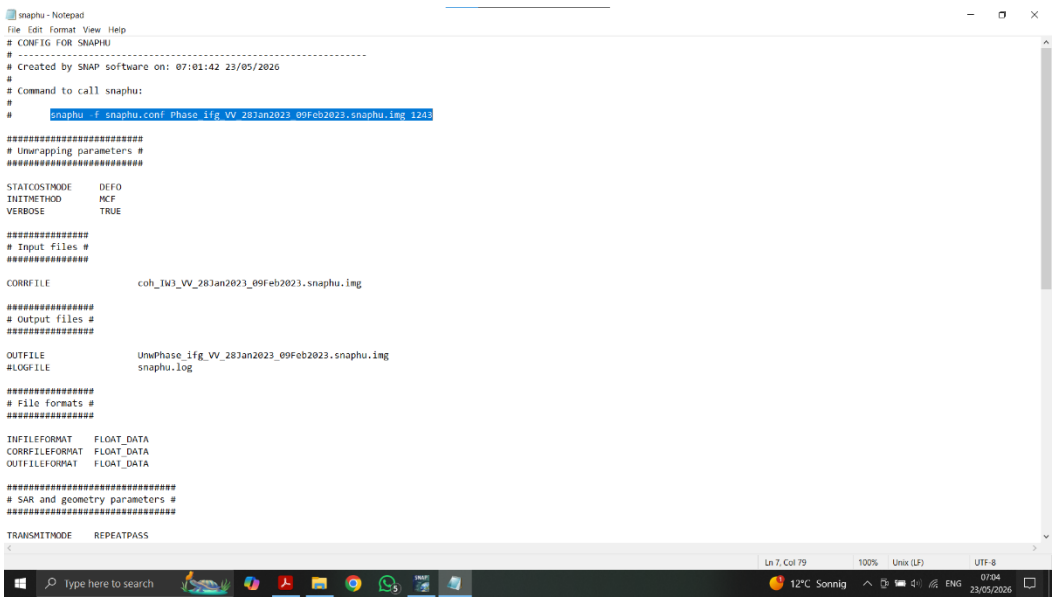


Figure 32. Adding hashtag, coherence file name, copying section and saving the changes in "snaphu" CONF file (Author)

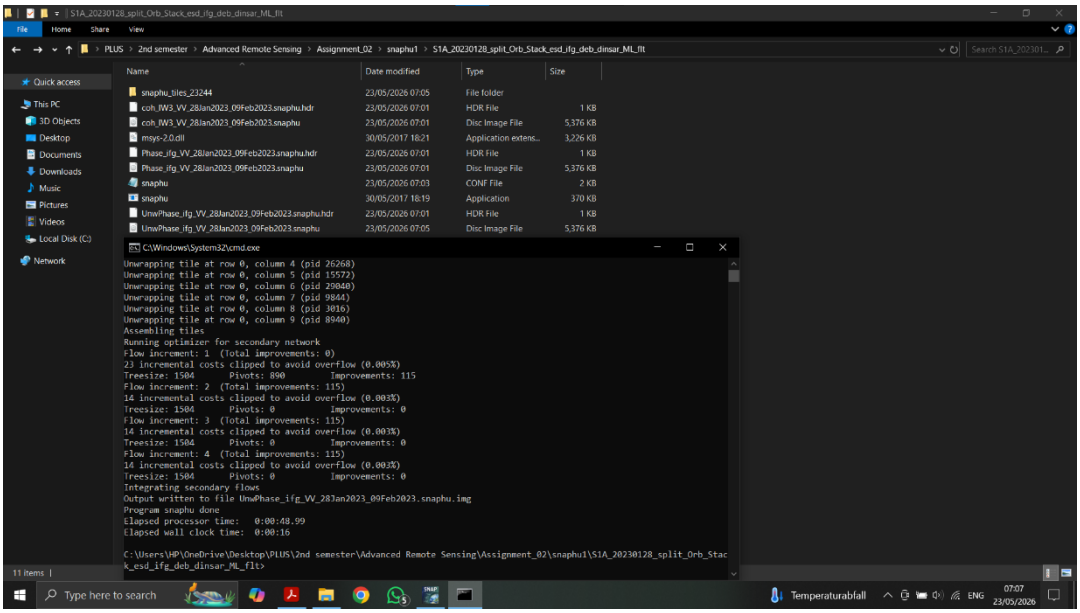


Figure 33. Successful unwrapping of the phase (Author)

I) Snaphu Import

After successful unwrap of the phase, it was necessary to import the corresponding file into SNAP. For this, it was selected “Radar”, “Interferometric, “Unwrapping” and then “Snaphu Import”. In the “1-Read-Phase” section, the last created differential interferogram was selected as “Source Product”. In the “2-Read-Unwrapped-Phase” section, the corresponding “UnwPhase” file was selected as “Source Product”. In the “4-Write” section, “un” was added to the output file product name.

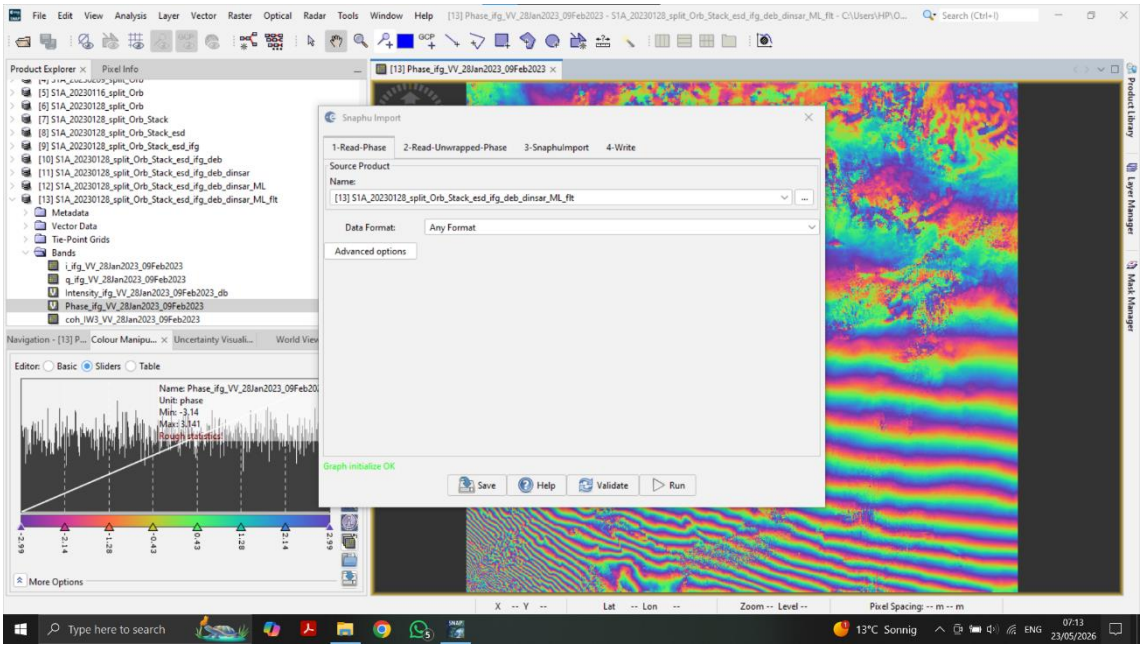


Figure 34. Selecting the last created differential interferogram as input in "1-Read-Phase" (Author)

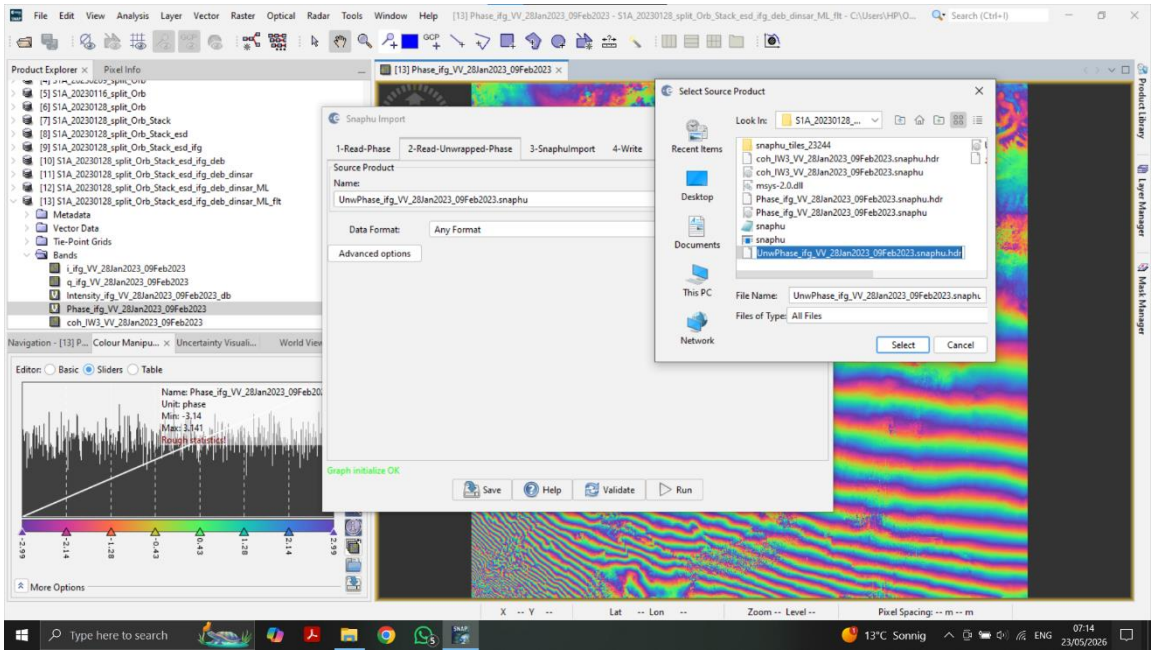


Figure 35. Selecting the corresponding file as input in "2-Read-Unwrapped-Phase" (Author)

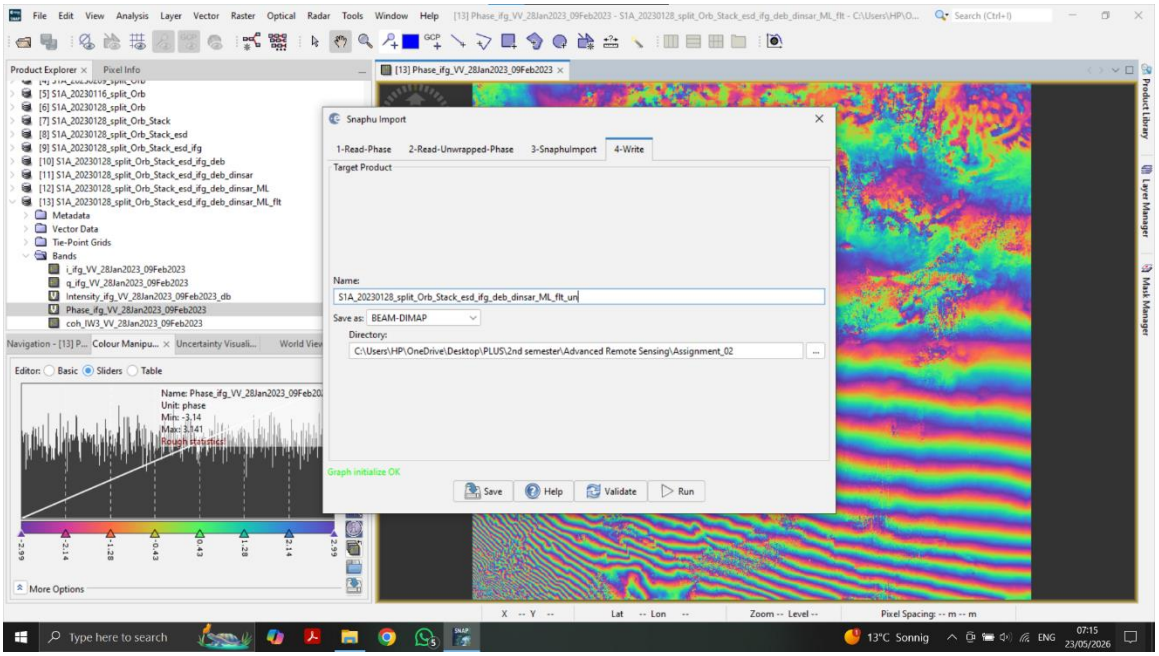


Figure 36. Adding "un" to the target product name in "4-Write" (Auhor)

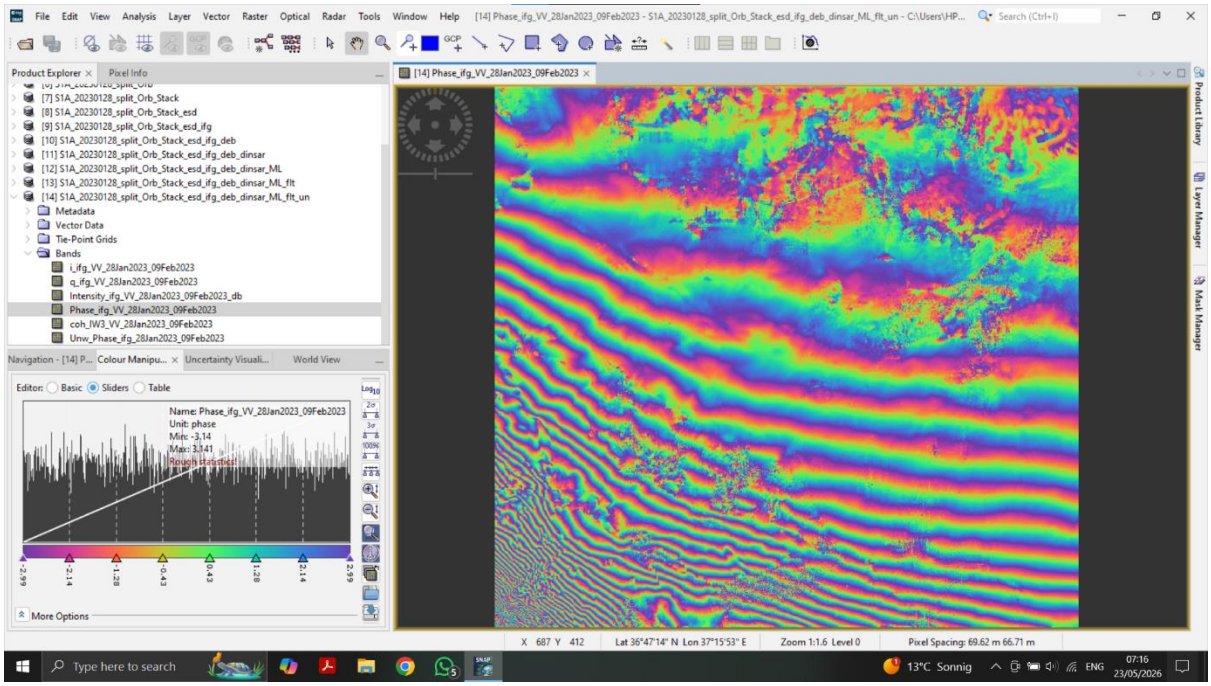


Figure 37. Representation of the unwrapped interferometric phase (Author)

m) Displacement Map Generation

Next step was to generate the diplacement map using the unwrapped phase product. It was selected “Radar”, “Interferometric”, “Products” and then “Phase to Displacement”. The input settings and paremters were left as they were.

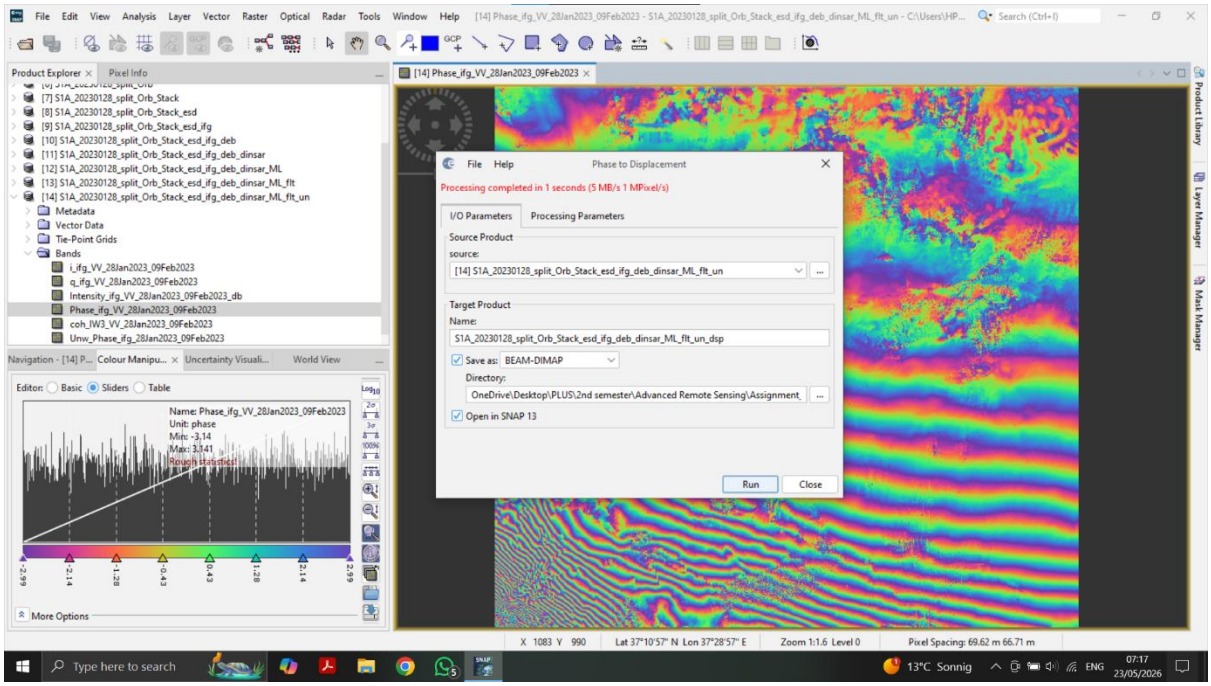


Figure 38. Generating the displacement map by performing "Phase to Displacement" (Author)

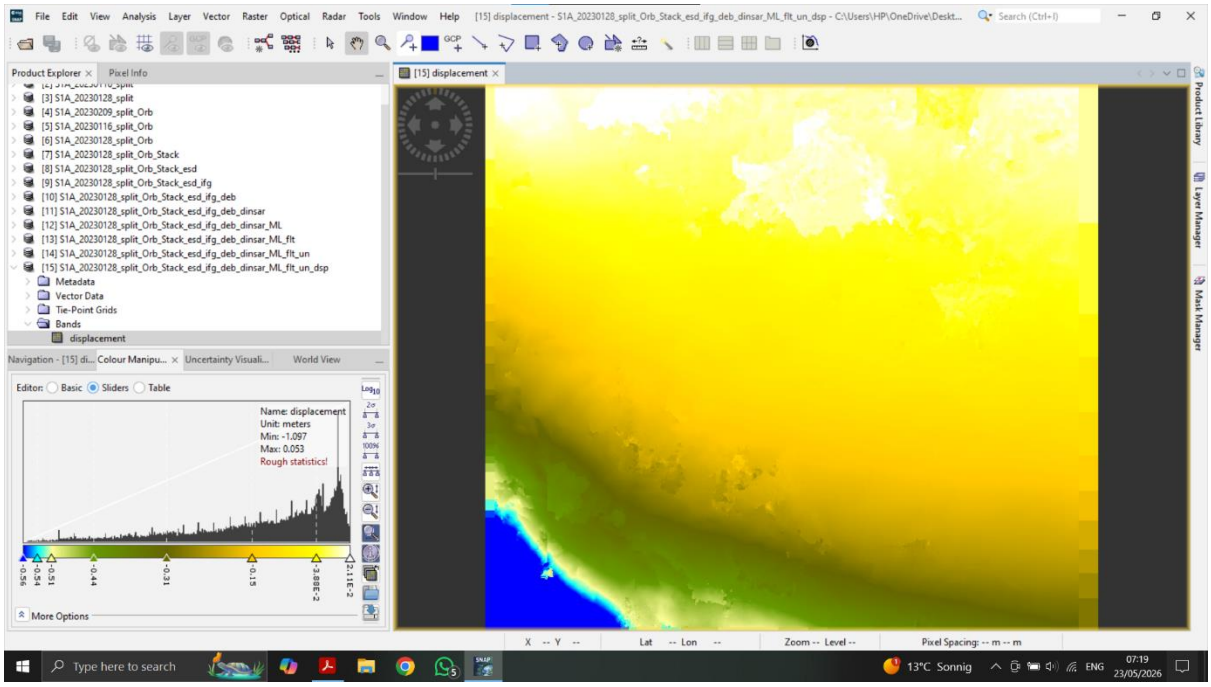


Figure 39. The resulting displacement map (Author)

n) Terrain Correction

The last step was to georeference the generated displacement map and remove the distortions caused by topography, to enable consistent analysis of the map. It was selected “Radar”, “Geometric”, “Terraing Correction” and then “Range-Dopler Terrain Correction”. The displacement map was selected as input and the DEM was changed

to “SRTM 1Sec HGT (Auto Download)”. Other settings and parameters were left as they were.

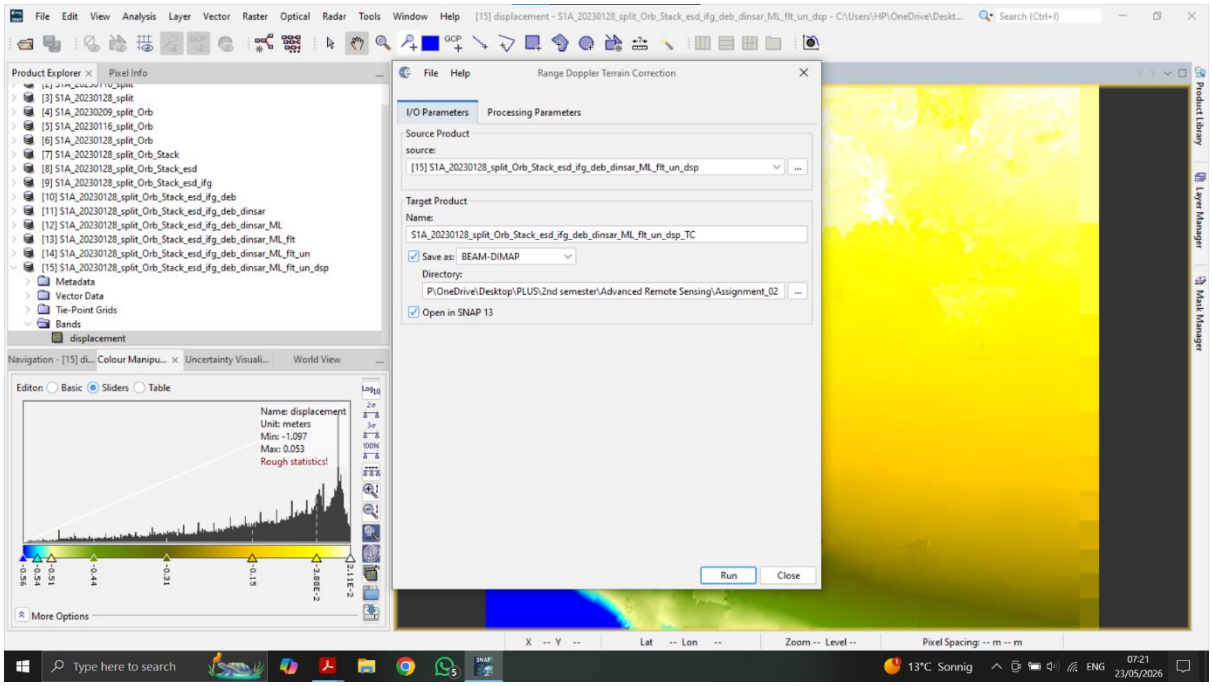


Figure 40. Selecting the displacement map as the input (Author)

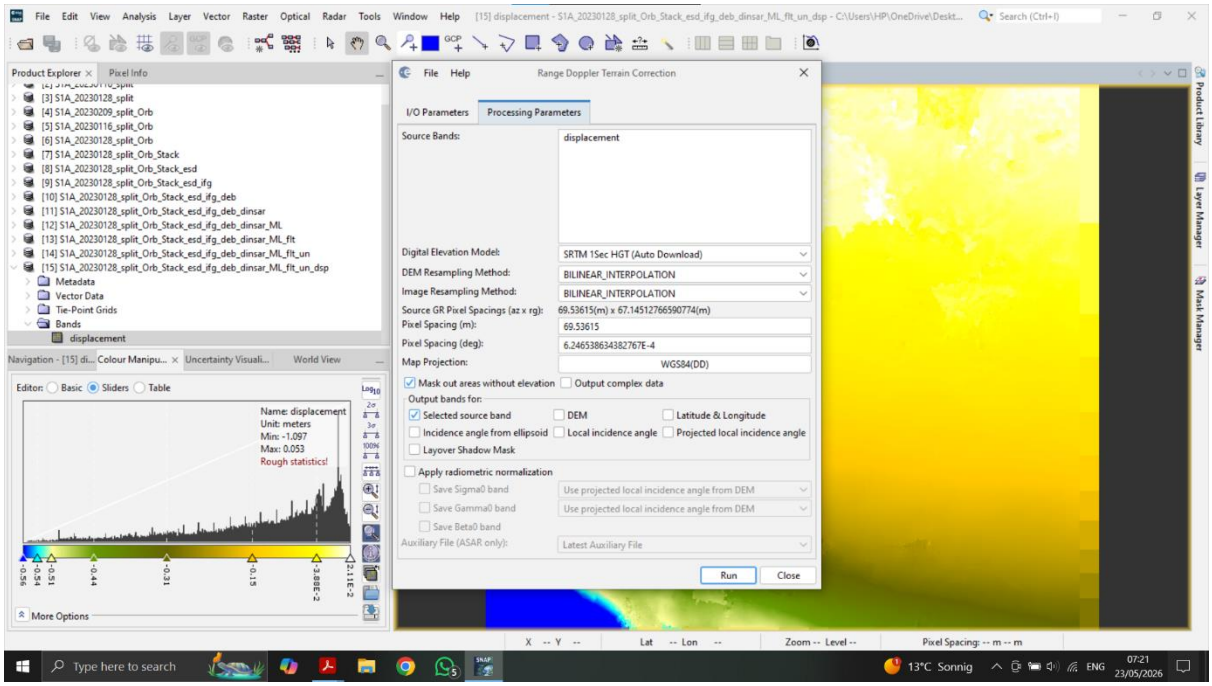


Figure 41. Changing DEM to "SRTM 1Sec HGT" (Author)

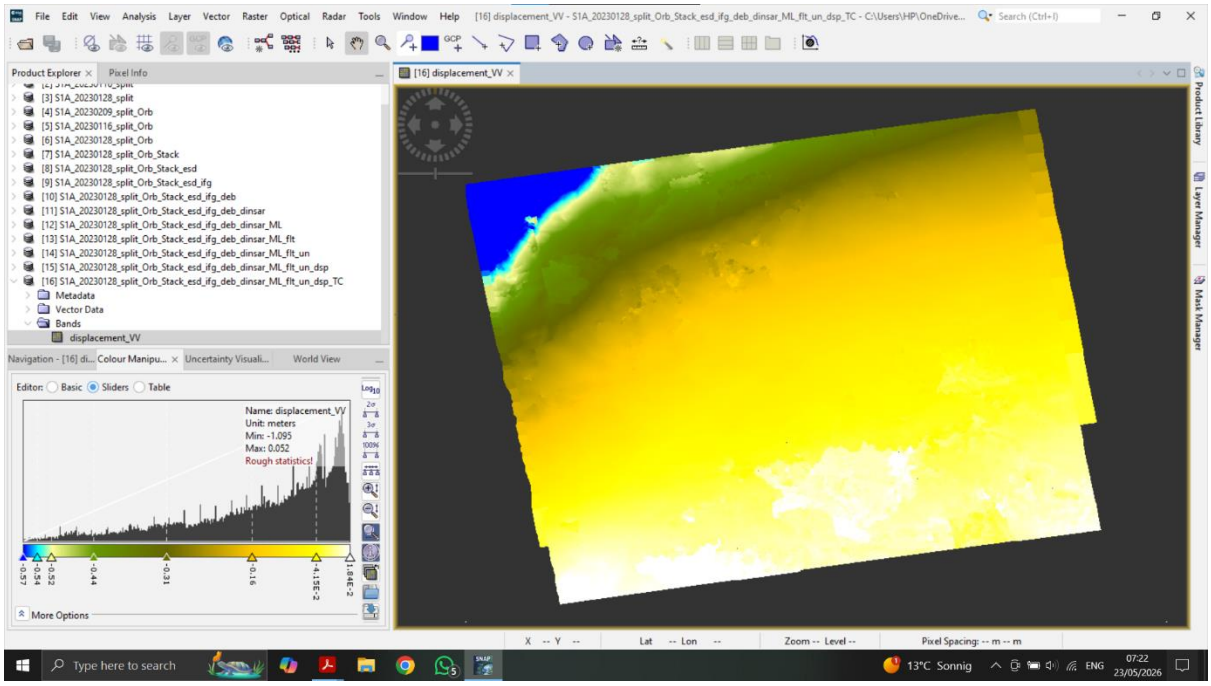


Figure 42. The resulting georeferenced and corrected displacement map (Author)

According to the final displacement map and sliders, the min value of -1.095 m is presented with blue color and can be observed in upper left corner. This is the area that had the most dense fringes in the interferometric phase representation. This means that significant displacement changes have occurred in relation to space, for example, displacement changes from -1.095 m to -0.080 m across smaller area. The max value of 0.052 m is presented with brighter colors (tendency towards white color) and can be observed in lower parts of the displacement map. This is the area that had less fringe patterns, that is, more irregularities in the interferometric phase representation. The positive and negative values mean that the displacement had different directions. As a conclusion, it can be said that earthquake which happened on 6th February 2023 had serious surface displacement, as well as fatal consequences.

After this, the displacement map was exported as a .tif file and loaded into ArcGIS Pro to observe the surface deformation in relation to the epicenter. The epicenter coordinates that were found at the beginning were loaded as a .csv file and converted using “XY Table To Point” tool.

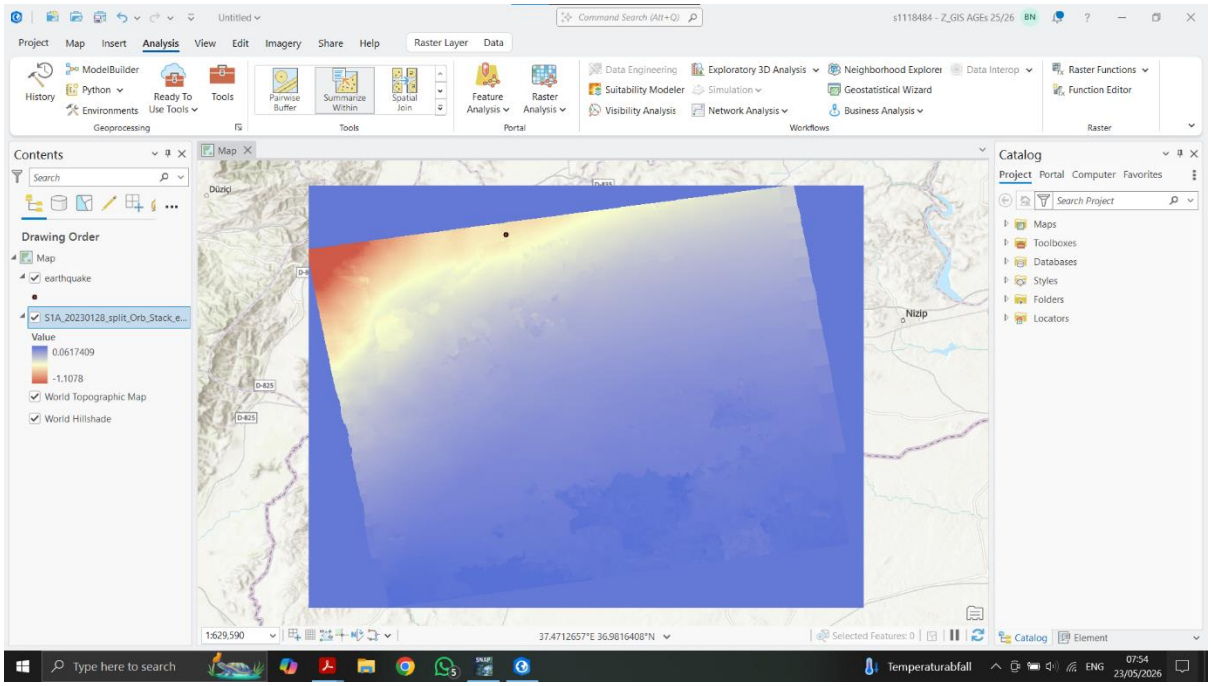


Figure 43. Representation of the displacement map together with epicenter in ArcGIS Pro (Author)

5. Topographic Interferogram & DEM

The technical part of this task was done using SNAP open source sofware developed by European Space Agency (ESA), same as for the creation of differential interferogram and displacement map. For the topographic interferogram, the topographic phase was not removed, since the goal was to observe the differences in phase caused by terrain elevation in AOI before the earthquake. Satellite images acquired on 16th January and 28th January (both before the earthquake) were used as the input datasets.

a) Loading Sentinel-1 images (Data Preparation)

Two corresponding datasets (together with the one that was used for the differential interferogram) were imported into SNAP using the “Open Product” option.

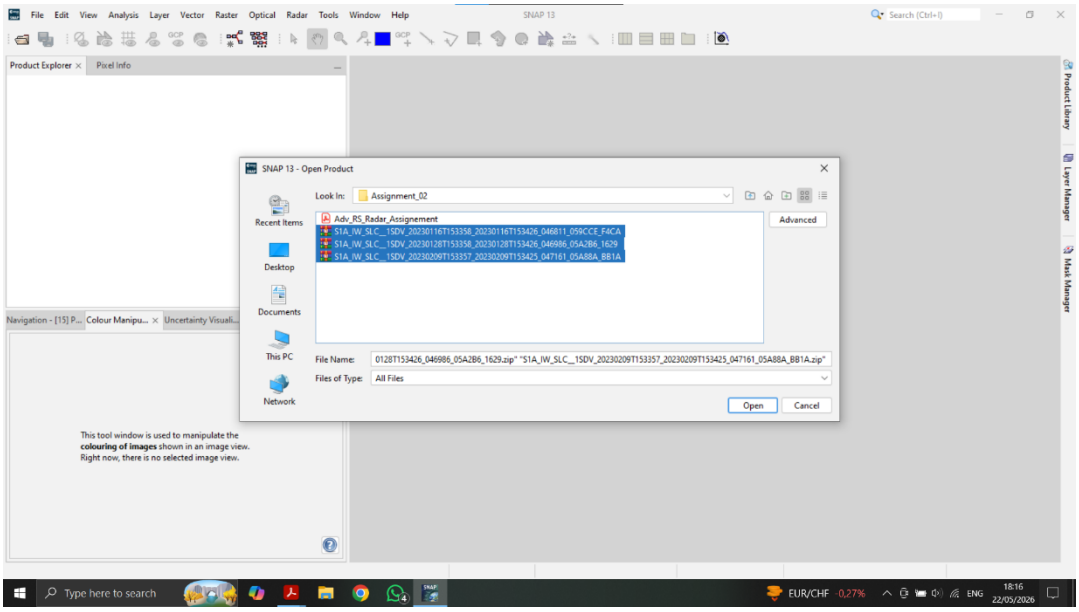


Figure 44. Importing required datasets into SNAP (Author)

b) Splitting the Subswath (Data Preparation)

After that, it was required to pre-process the datasets that will be used for creating the topographic interferogram. First step was to choose the subswath and burts that include AOI. It was selected “Radar”, “Sentinel-1 TOPS” and then “S-1 TOPS Split”. IW3 was selected for the subswath, VV was selected as the polarisation method, and the bursts were selected from 6 to 9. The target product names were shortened so they do not get too long by the end of process.

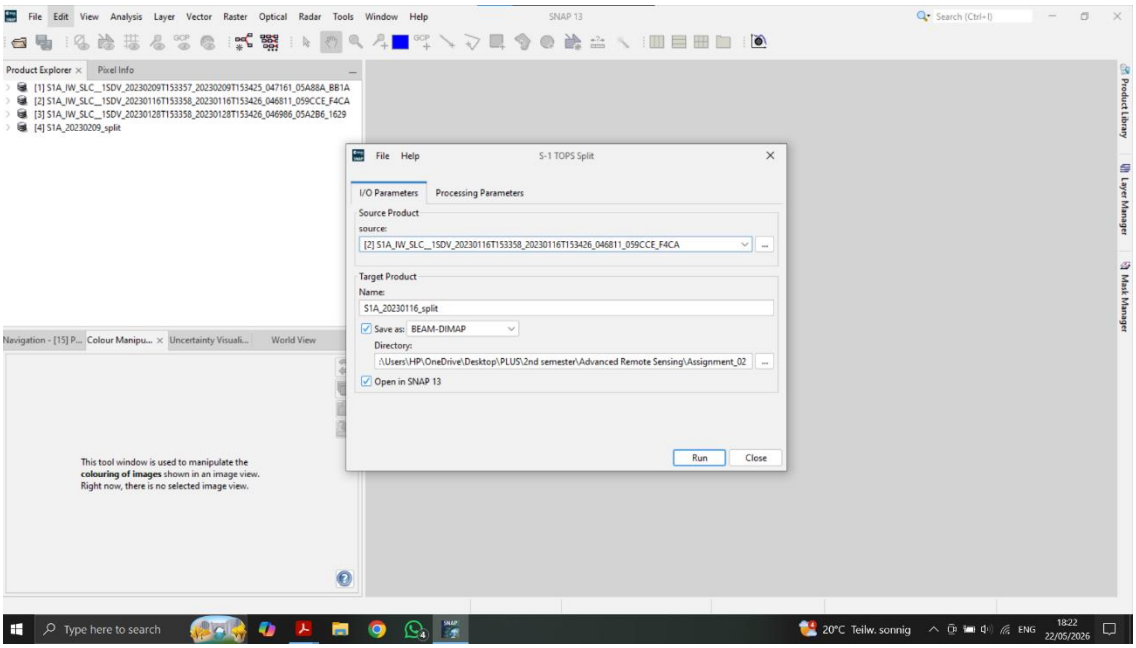


Figure 45. Chosing shorter name for the 16th January dataset while performing "S-1 TOPS Split" (Author)

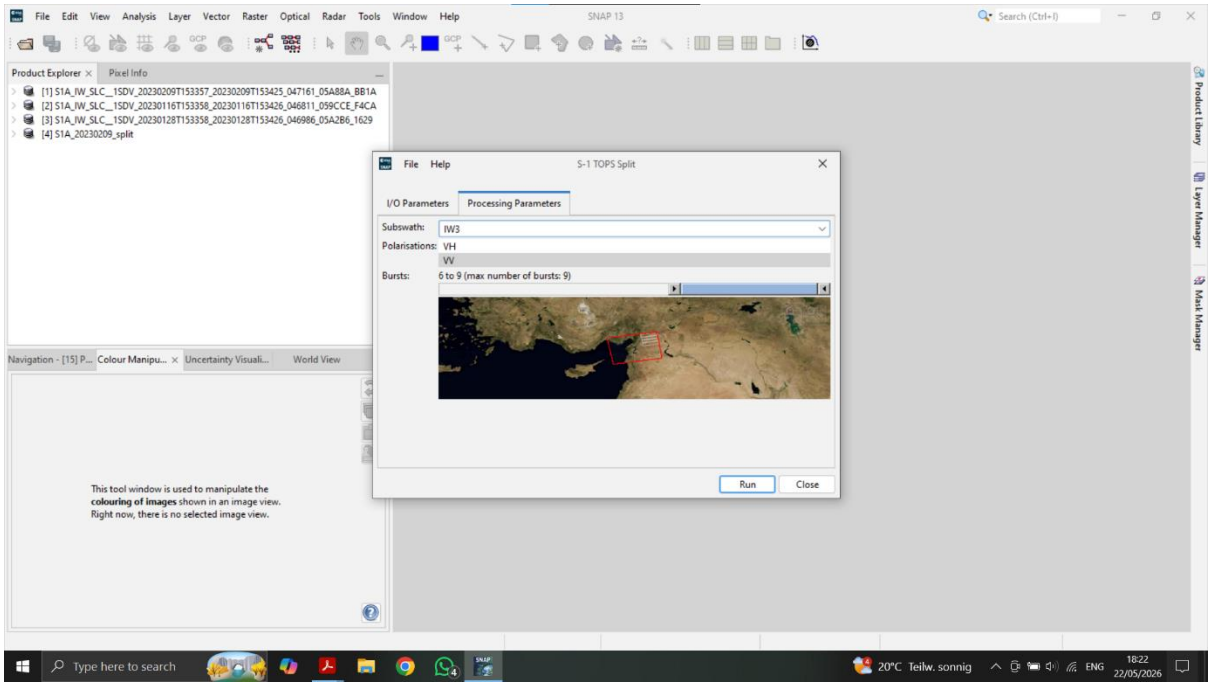


Figure 46. Selecting the subswath, polarisation method and burts that contain AOI for 16th January dataset (Author)

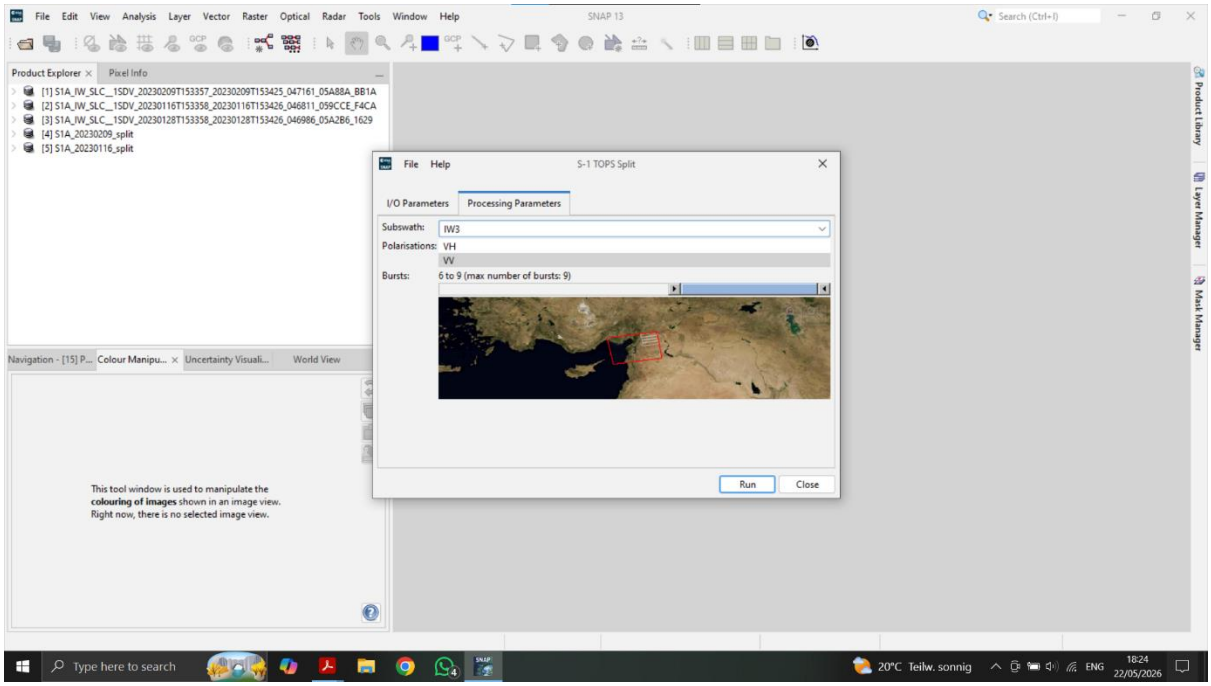


Figure 47. Selecting the subswath, polarisation method and burts that contain AOI for 28th January dataset (Author)

c) Applying Orbit File (Data Preparation)

The last step of data preparation was to correct the satellite positioning errors by selecting “Radar” and then “Apply Orbit File”. The processing parameteres were left as they were.

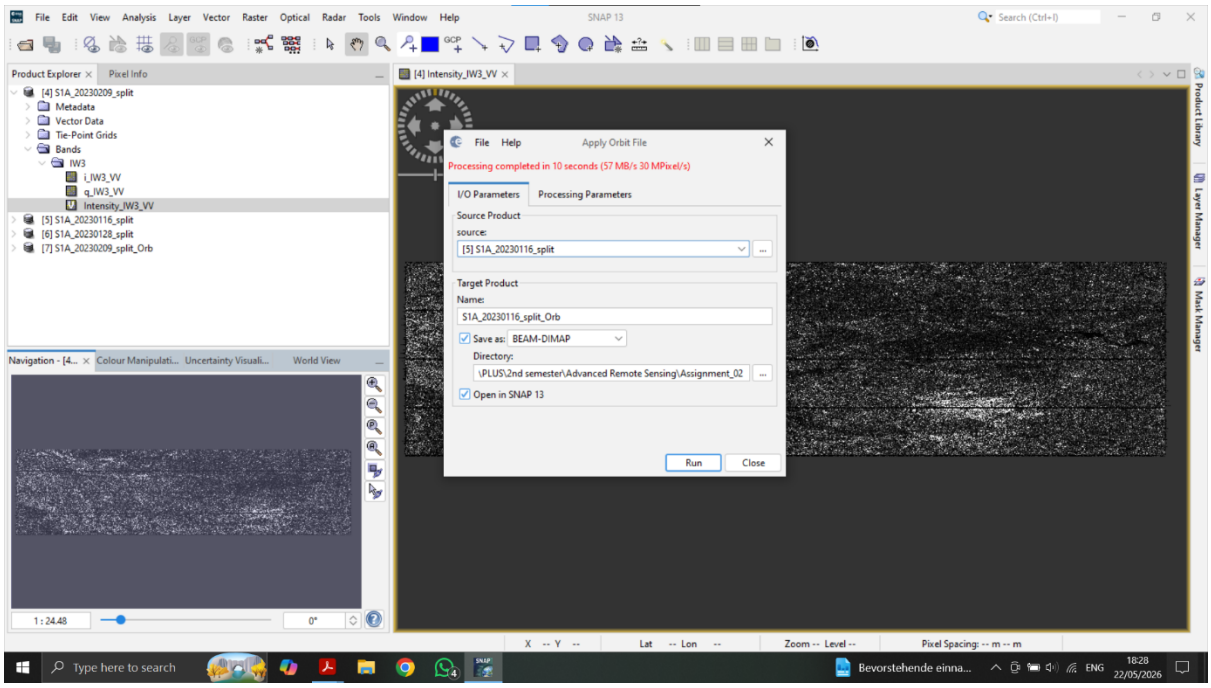


Figure 48. Performing "Apply Orbit File" for 16th January dataset (Author)

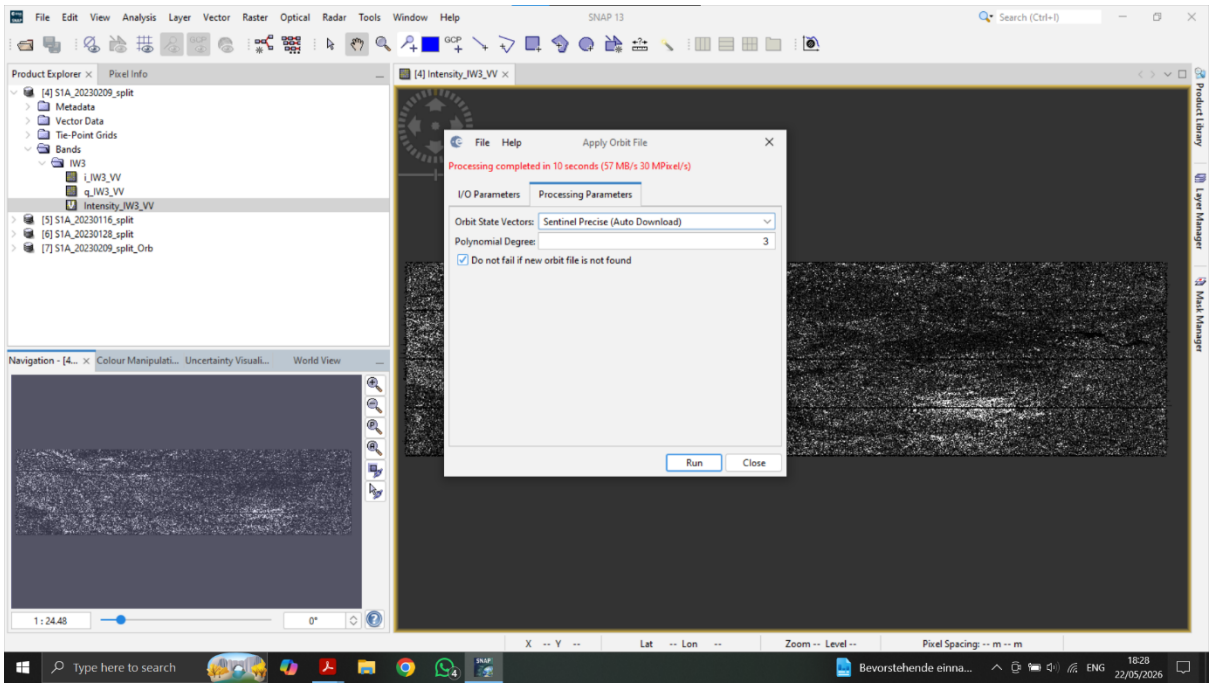


Figure 49. Checking processing parameters in "Apply Orbit File" for 16th January dataset (Author)

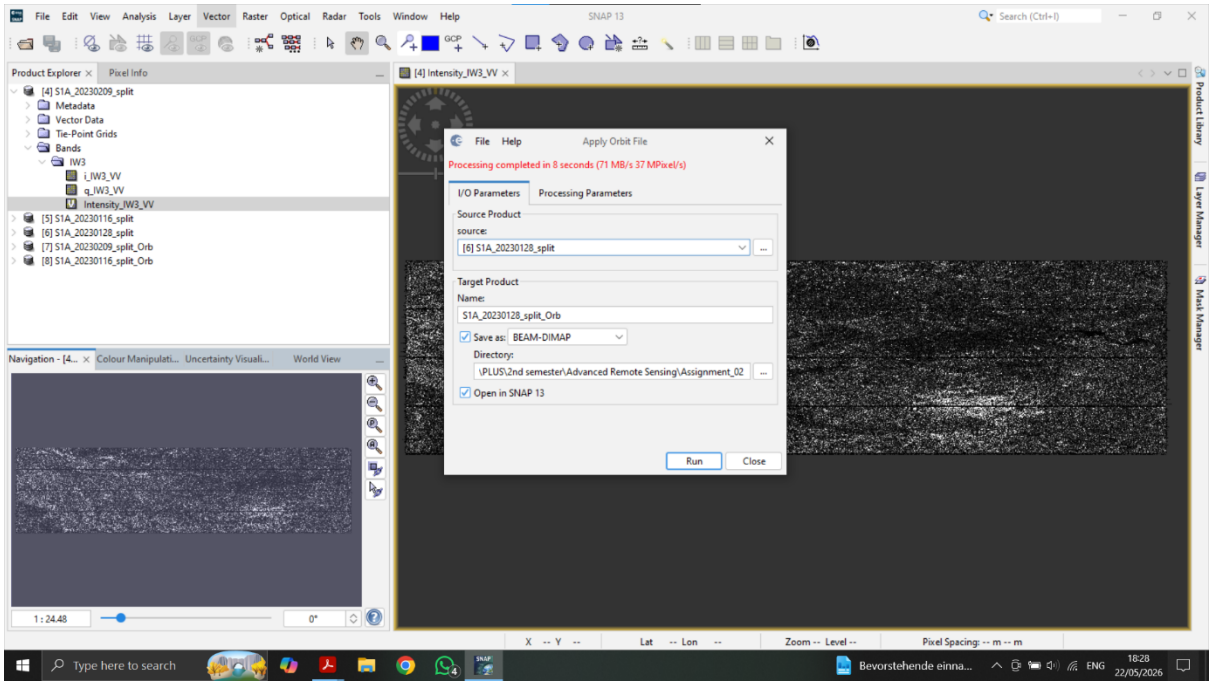


Figure 50. Performing "Apply Orbit File" for 28th January dataset (Author)

d) Back Geocoding (Coregistration)

In this part, it was necessary to align the mater and slave satellite images. For this was selected “Radar”, “Coregistration”, “S-1 TOPS Coregistration” and then “S-1 Back Geocoding”. Then, then the master and slave images were added, and the DEM was changed to “SRTM 1Sec HGT (Auto Download)”. The rest was left as it is.

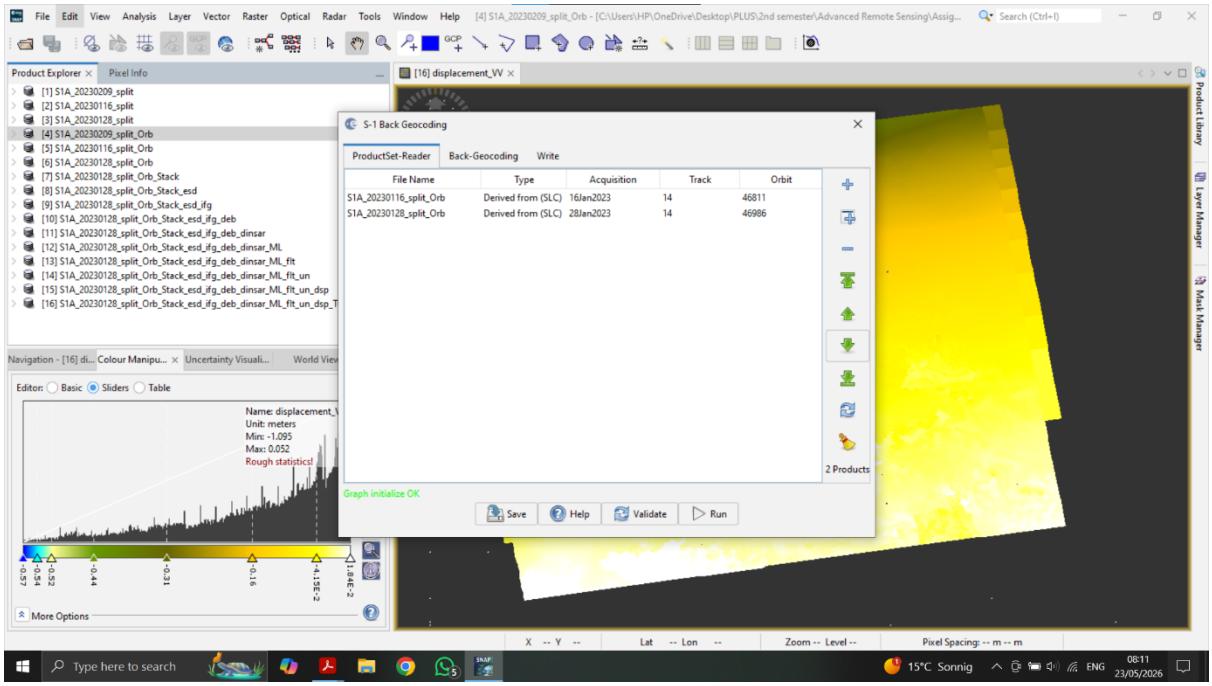


Figure 51. Adding the master and slave images in the "ProductSet-Reader" section (Author)

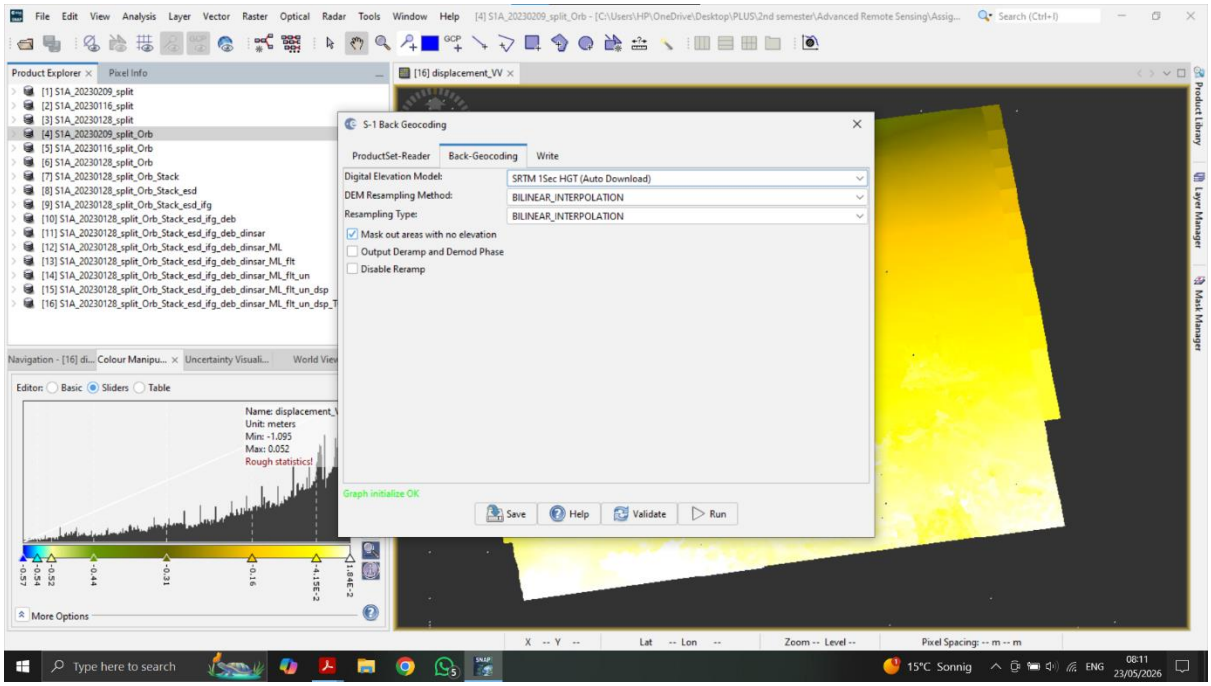


Figure 52. Changing DEM in the "Back-Geocoding" section (Author)

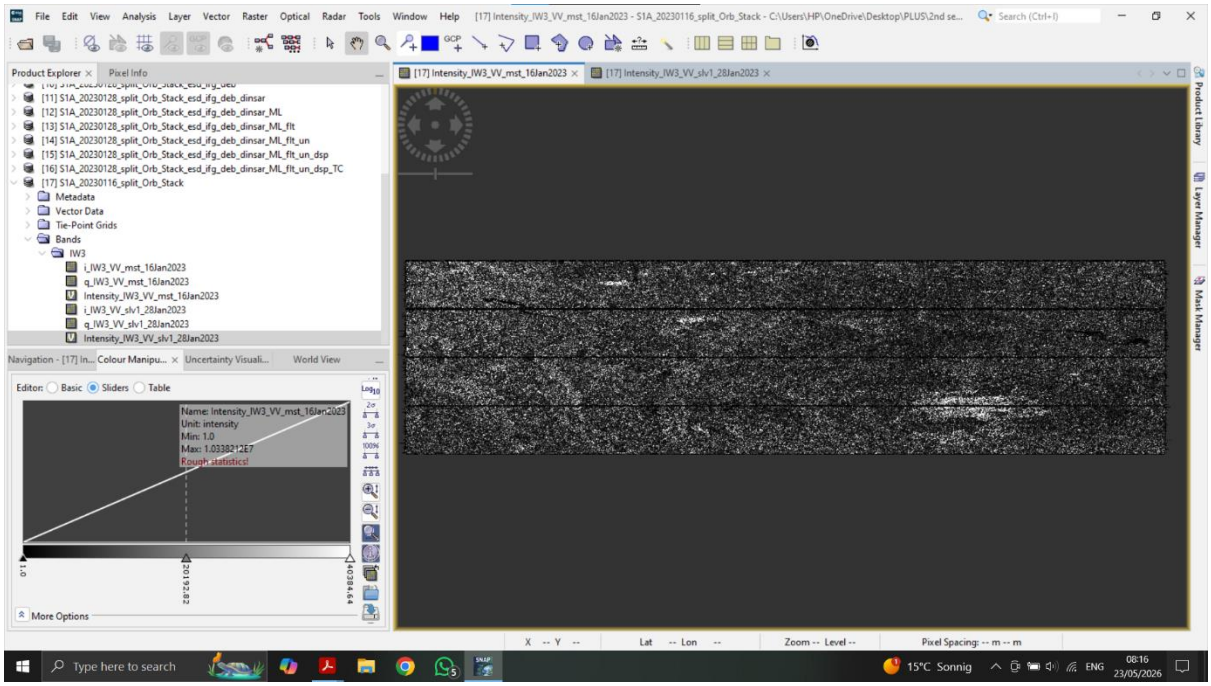


Figure 53. The outcome for 16th January dataset (Author)

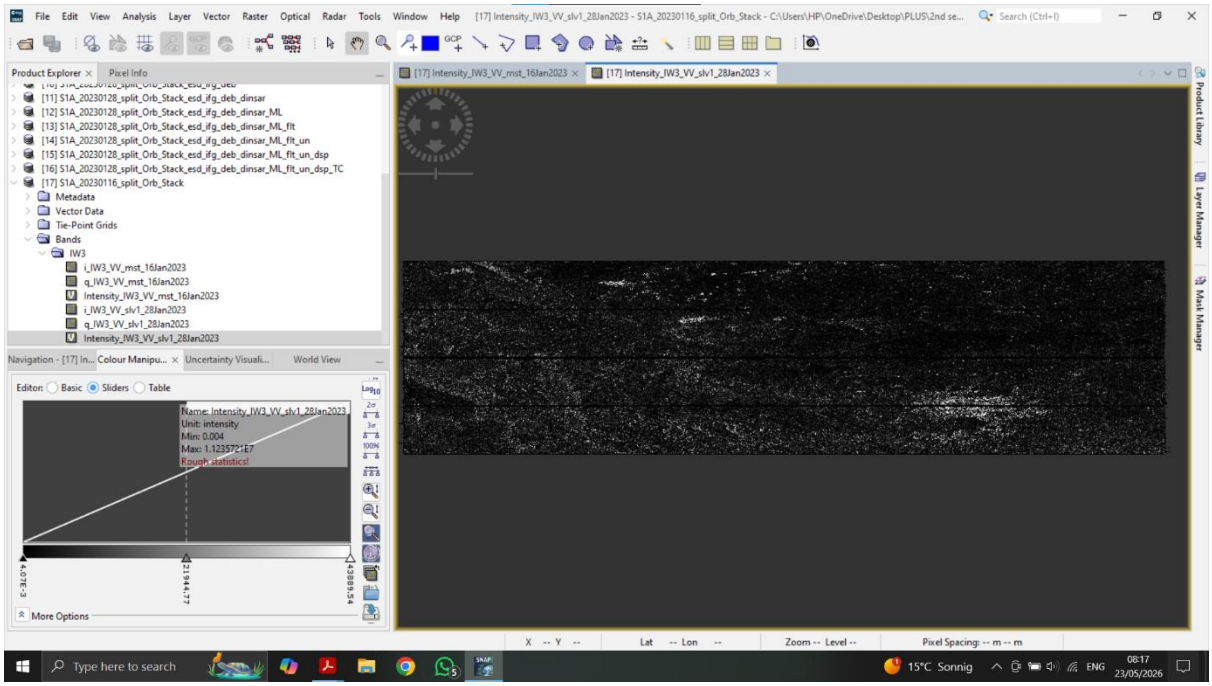


Figure 54. The outcome for 28th January dataset (Author)

e) Enhanced Spectral Diversity (Coregistration)

It was selected “Radar”, “Coregistration” and then “S-1 Enhanced Sprectral Diversity”. There was no need for changes in settings.

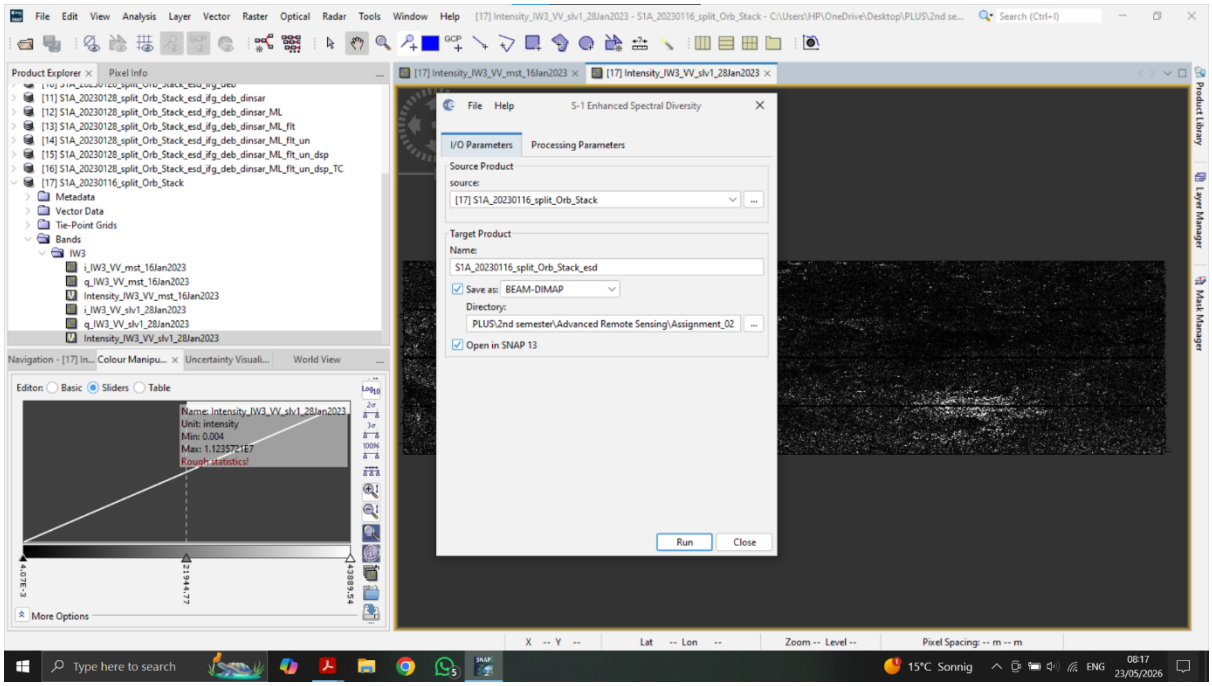


Figure 55. Performing "Enhanced Spectral Diversity" (Author)

f) Interferogram Formation (Topographic Interferogram Generation)

This step included generating topographic interferogram to later observe the differences in phase caused by terrain elevation. It was selected “Radar”, “Interferometric”, “Products” and then “Interferogram Formation”. For the processing parameters, “Coherence Range Window Size” was changed to 20, and “Coherence Azimuth Window Size” automatically changed to 5. The rest was left as it is.

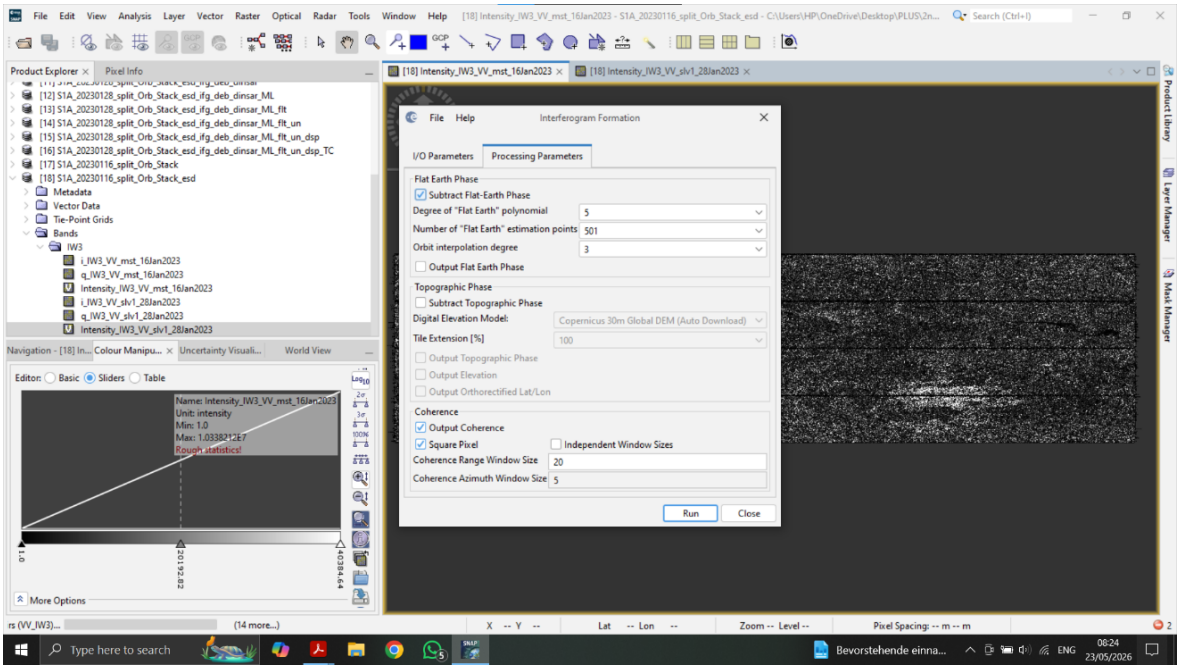


Figure 56. Changing the processing parameters for "Interferogram Formation" (Author)

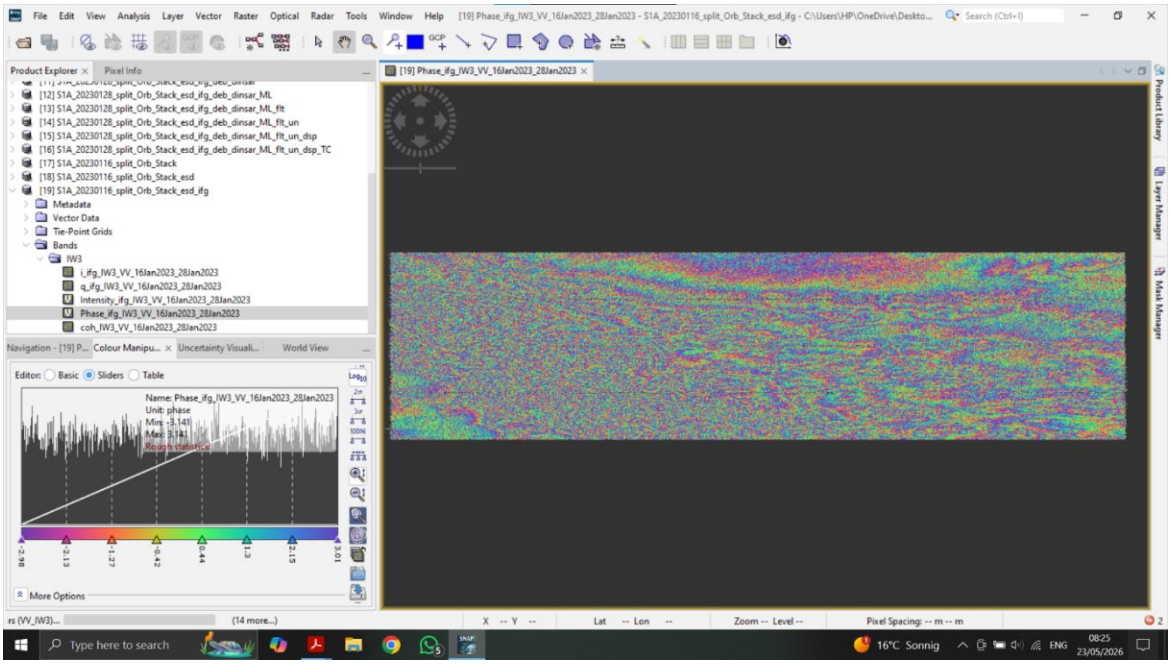


Figure 57. The resulting interferogram (Author)

g) TOPSAR Deburst

Next step was to remove the burst boundaries which resulted in merged interferogram. For this step was selected “Radar”, “Sentinel-1 TOPS” and then “S-1 TOPS Deburst”. The settings were left as they were.

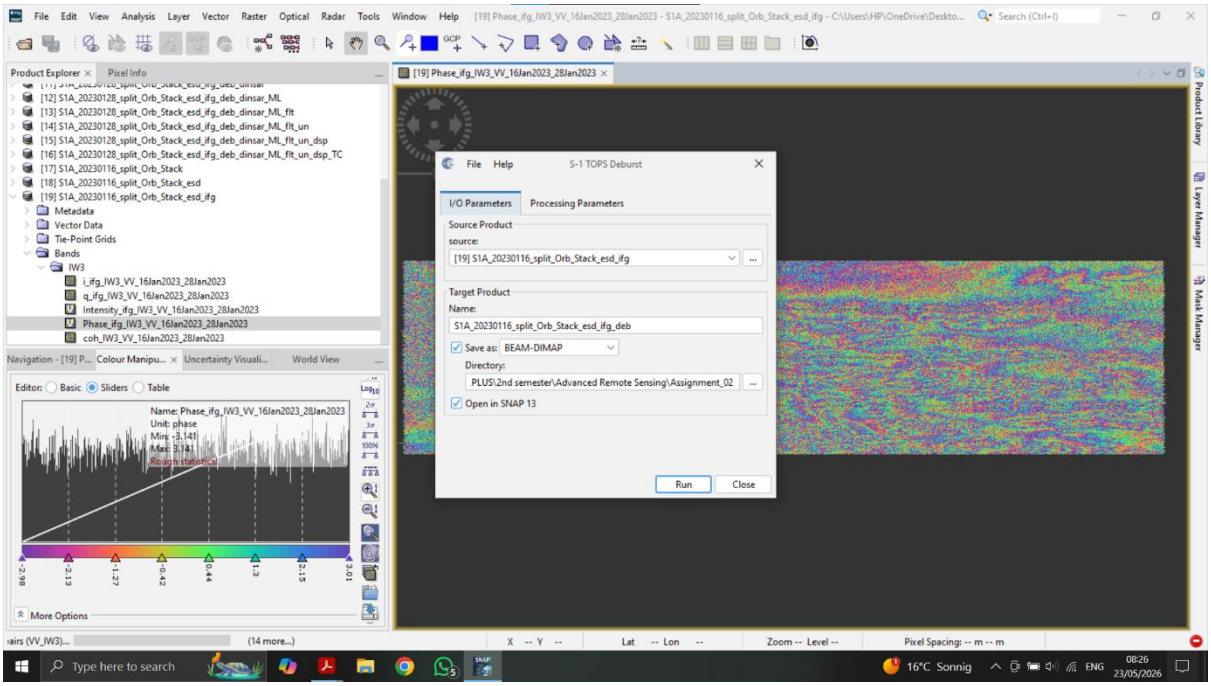


Figure 58. Peforming "S-1 TOPS Deburst" (Author)

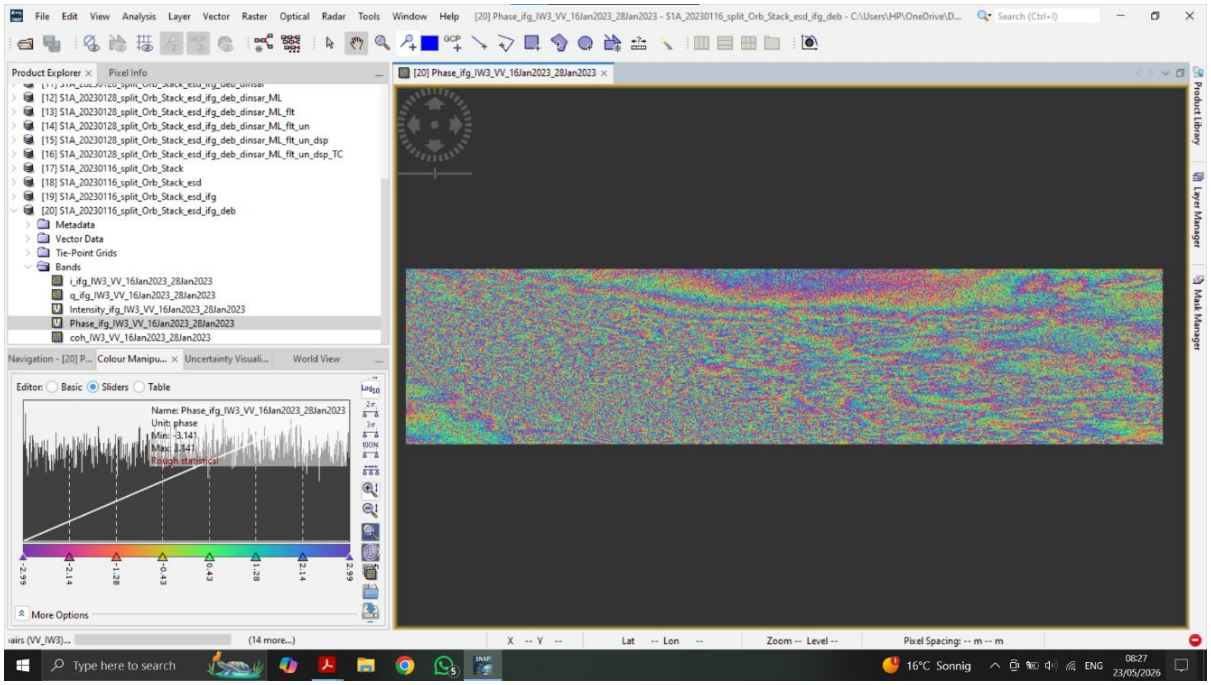


Figure 59. The resulting merged interferogram (Author)

h) Multilooking

In this step it was required to make the topographic interferogram more visually interpretable by performing “Multilooking”, which reduces noise and makes pixels squared. It was selected “Radar”, “SAR Utilities” and then “Multilooking”. In the “Processing Parameters” section, “Number of Range Looks” was changed to 20, and “Number of Azimuth Looks” automatically changed to 5. The input was topographic interferogram from the last step. The rest was left as it is.

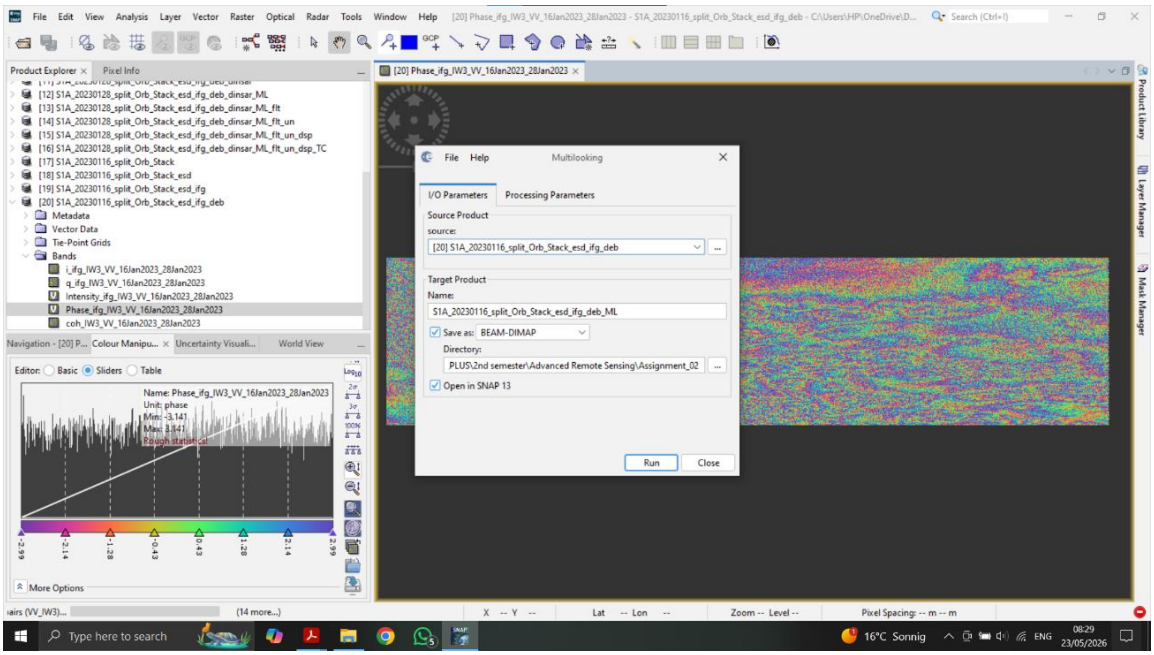


Figure 60. Selecting the topographic interferogram from the last step as input (Author)

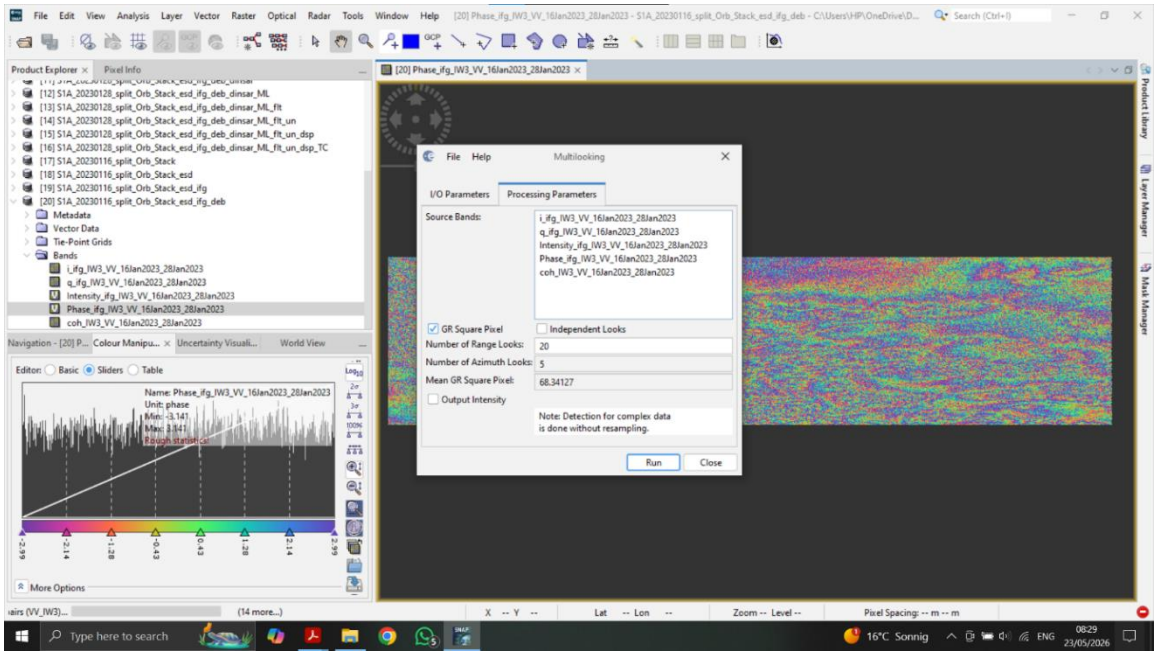


Figure 61. Changing the processing parameters for "Multilooking" (Author)

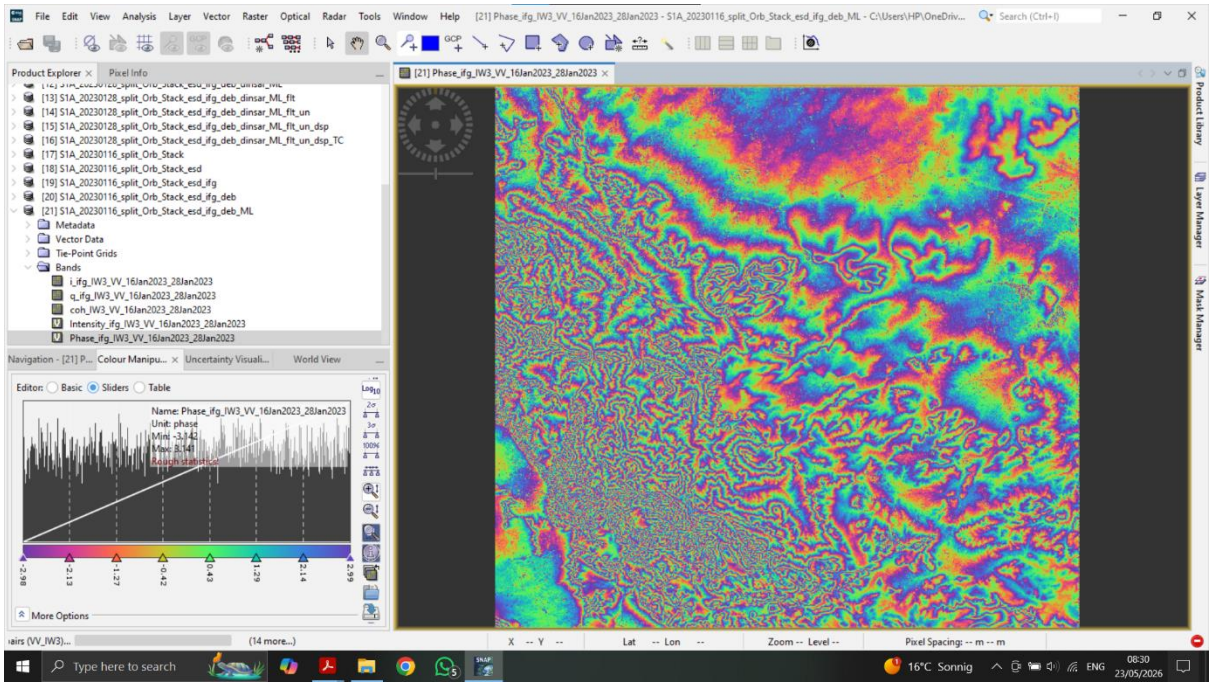


Figure 62. The resulting topographic interferogram after performing "Multilooking" (Author)

i) Goldstein Phase Filtering

This was the last step of processing the topographic interferogram (before Snaphu export), and its purpose was to improve the phase quality (reduce noise) to ensure accurate unwrapping and elevation calculation. “Coherence Threshold” was changed to 0.8 in the processing parameters section. The rest was left as it is.

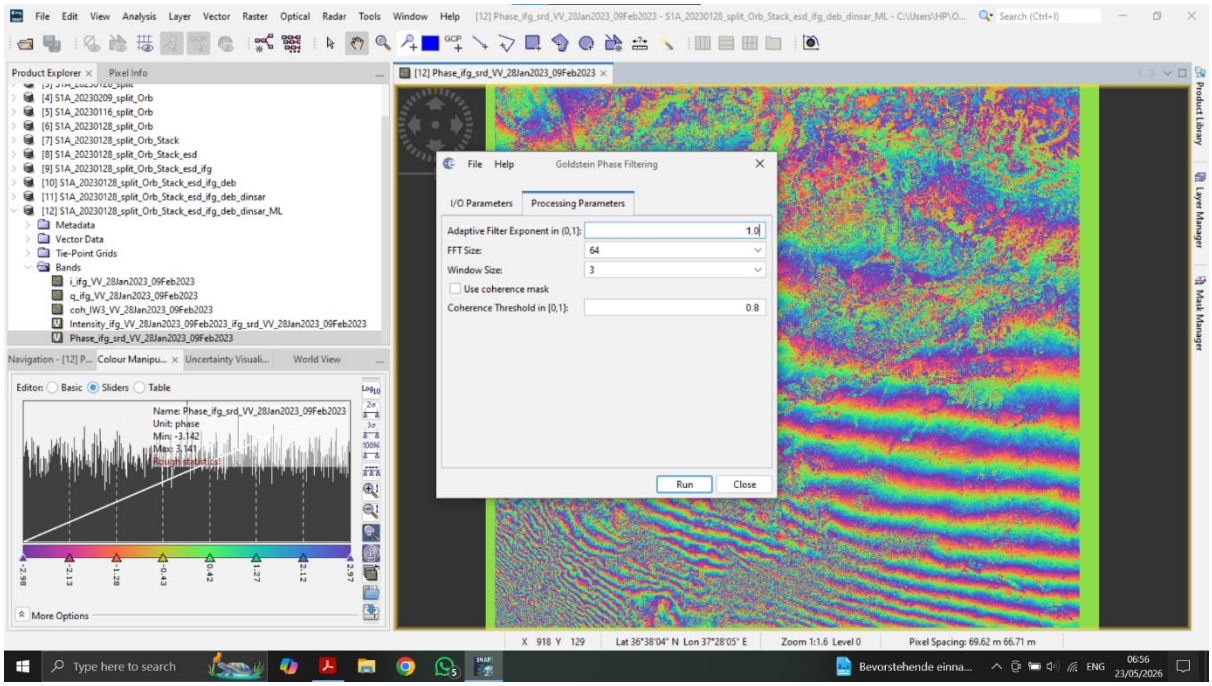


Figure 63. Changing the "Coherence Threshold" value to 0.8 (Author)

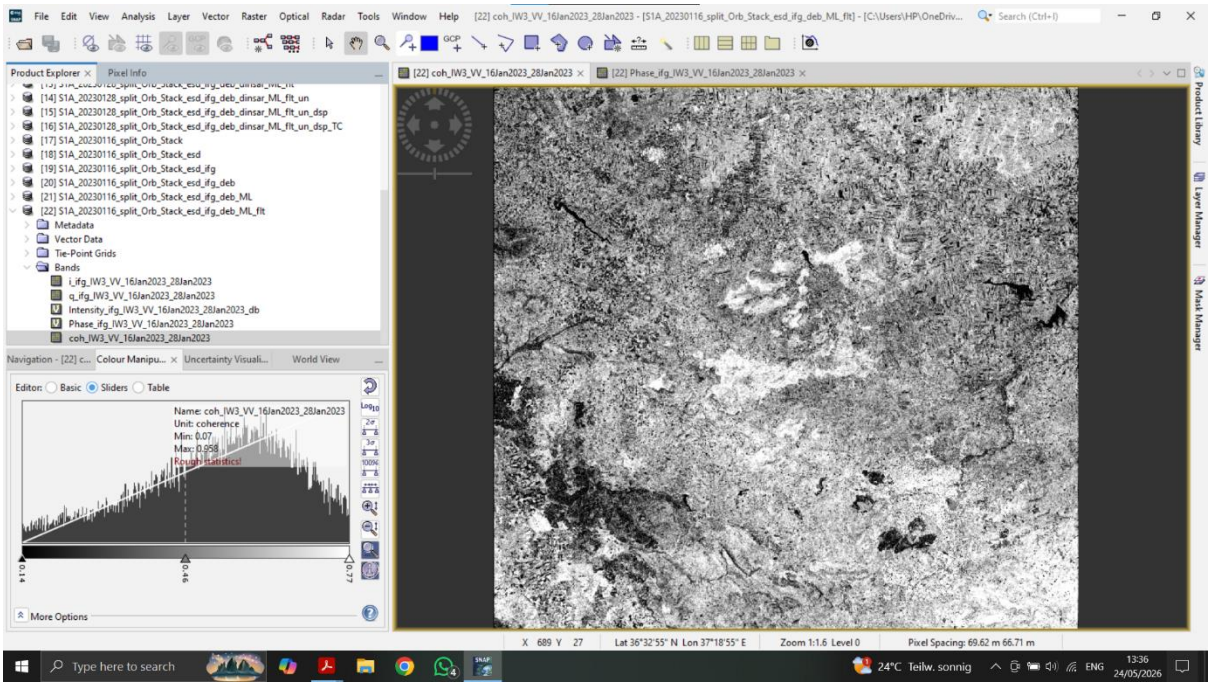


Figure 64. The resulting interferometric coherence after performing "Goldstein Phase Filtering" (Author)

The interferometric coherence measures the quality of the interferometric pair. Areas represented with brighter colors mean that phase has better quality, whereas areas represented with darker colors mean that phase has lower quality. The values range from 0.07-0.958 according to the statistics section. The central (white spots) and lower right part (excluding the dark spots) have more brighter colors. Dark areas can be observed in lower left part of the interferometric coherence representation. Upper part of the representation has mixed colors. Overall phase quality is more on the correct side according to the sliders. Compared to the differential interferometric coherence representation, the topographic coherence appears more brighter, meaning that the quality of the phase between the topographic pair is better.

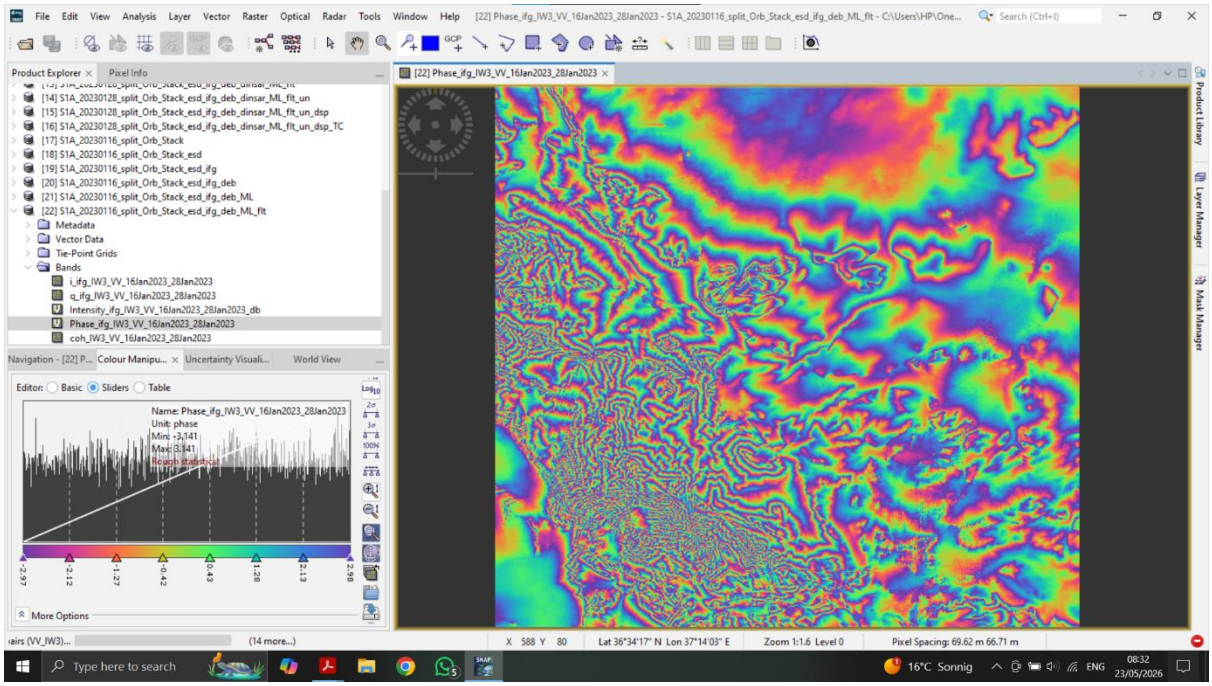


Figure 65. The resulting interferometric phase after performing "Goldstein Phase Filtering" (Author)

As it can be observed in (topographic) interferometric phase representation, fringe patters and their spatial continuity are mostly irregular. Somewhat spatial continuity can be seen in lower left to central left parts. Fringe density is the highest in the lower left and central left parts of the representation (excluding lower left corner), meaning that these areas have significant elevation change across space (may be steep mountain). On the other hand, the fringe density it the lowest in lower left corner and upper right part, which means that these areas have less elevation changes across space. Compared to the differential interferometric phase representation, the topographic interferometric phase has less spatial continuity, but the density appears somewhat similar depending on the observed part. For example, fringe density is the highest in the lower left corner in differential phase representation, while this part is not dense in topographic phase representation. This means that surface deformation has ocurred across smaller space which does not have significant elevation changes (for example not steep area).

j) Snaphu Export

The next step of the analysis was to create DEM. In order to perform this, it was necessary to unwrap the phase using Snaphu to obtain continuous elevation measurements. First step was to select “Radar”, “Interferometric”, “Unwrapping” and then “Snaphu Export”. For the “Source product”, the topographic interferogram created

in the last step was selected. In the settings, the target folder was defined, “Number of Tile Rows” and “Number of Tile Columns” were set to 1, “Number of Processors” was set to 4, “Row Overlap” and “Column Overlap” were set to 0, and “Tile Cost Threshold” was set to 500. Other settings were not changed.

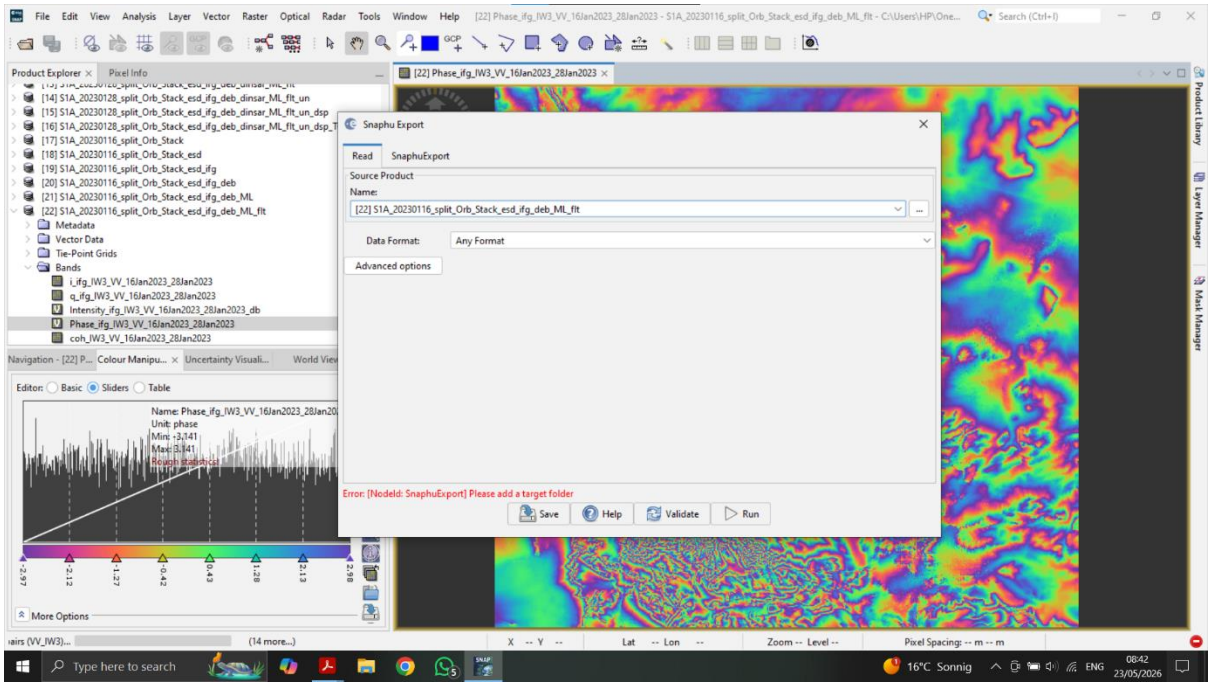


Figure 66. Selecting the topographic interferogram from the last step as "Source Product" (Author)

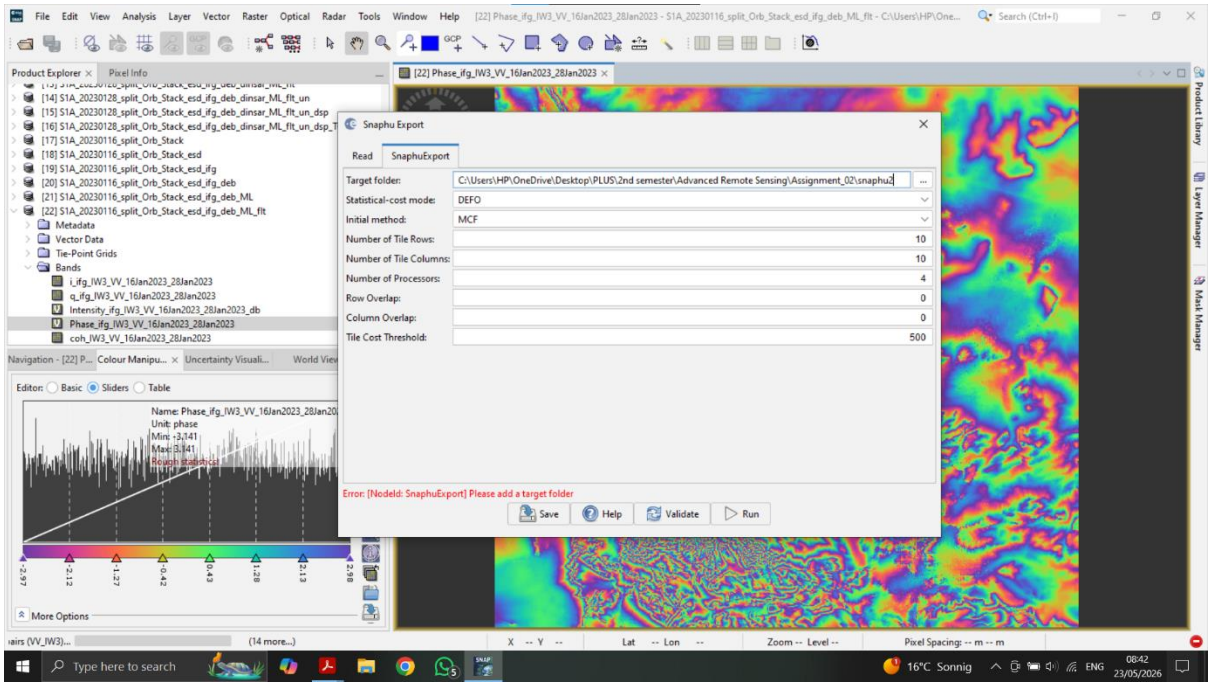


Figure 67. Defining the settings for "Snaphu Export" (Author)

After that, the “snaphu” application and “msys-2.0.dll” application extension were placed in the exported “snaphu2” folder. Then “snaphu” CONF file was opened and hashtag was addded before “LOGFILE”, and the coherence file name was added to the “CORRFILE” line. Section “snaphu -f snaphu.conf Phase_ifg_IW3_VV_16Jan2023_28Jan2023.snaphu.img 1243” was copied and the changes were saved. Next, the copied section was pasted in the command line and executed. When the unwrapping finished, a new “UnwPhase” fille occurred.

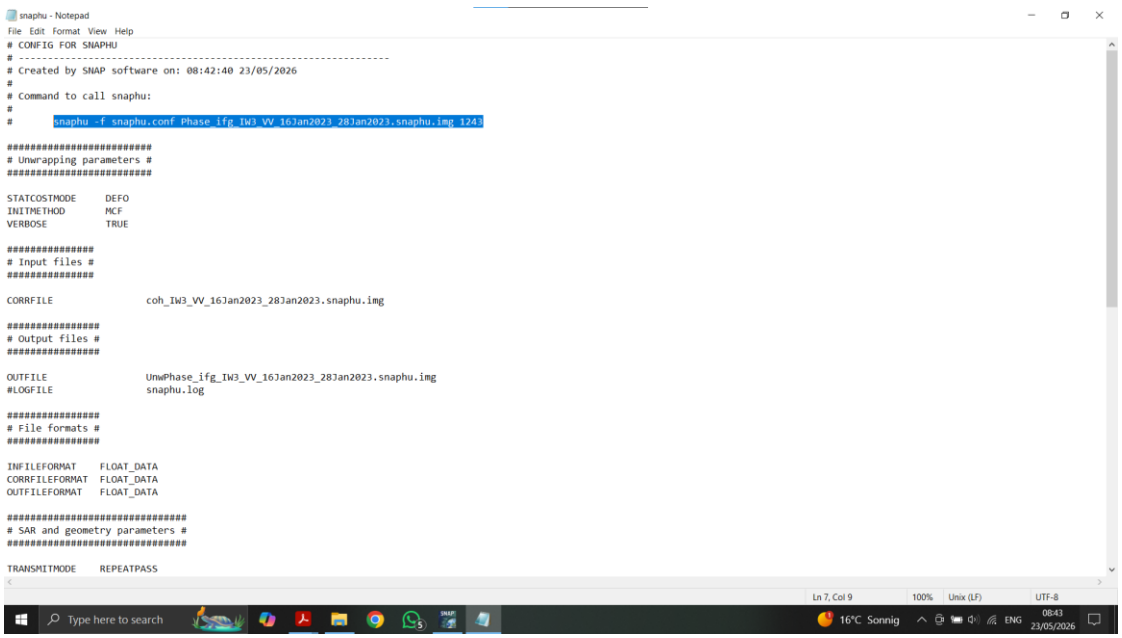


Figure 68. Adding hastag, coherence file name, copying section and saving the changes in "snaphu" CONF file (Author)

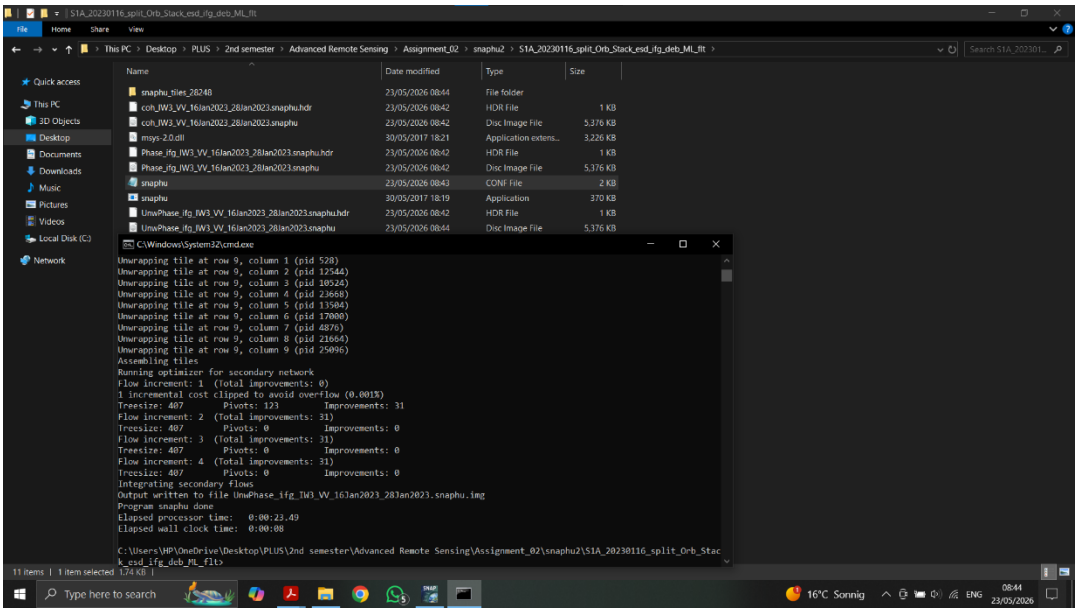


Figure 69. Successful unwrapping of the phase (Author)

k) Snaphu Import

After successful unwrap of the phase, it was necessary to import the corresponding file into SNAP. For this, it was selected “Radar”, “Interferometric”, “Unwrapping” and then “Snaphu Import”. In the “1-Read-Phase” section, the last created topographic interferogram was selected as “Source Product”. In the “2-Read-Unwrapped-Phase” section, the corresponding “UnwPhase” file was selected as “Source Product”. In the “4-Write” section, “un” was added to the output file product.

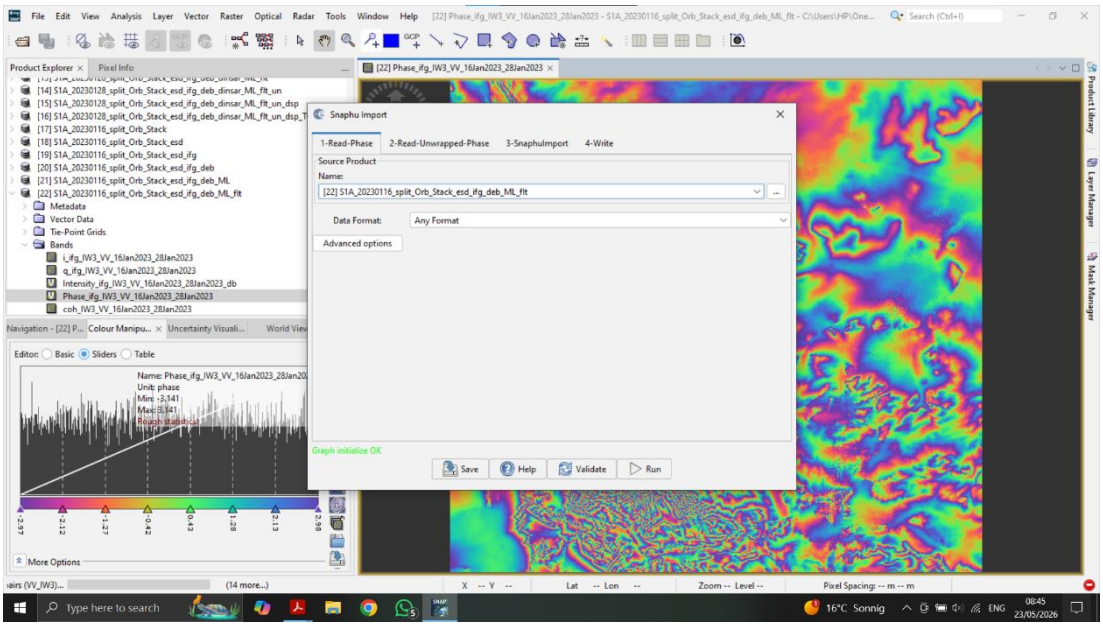


Figure 70. Selecting the last created topographic interferogram as input in "1-Read-Phase" (Author)

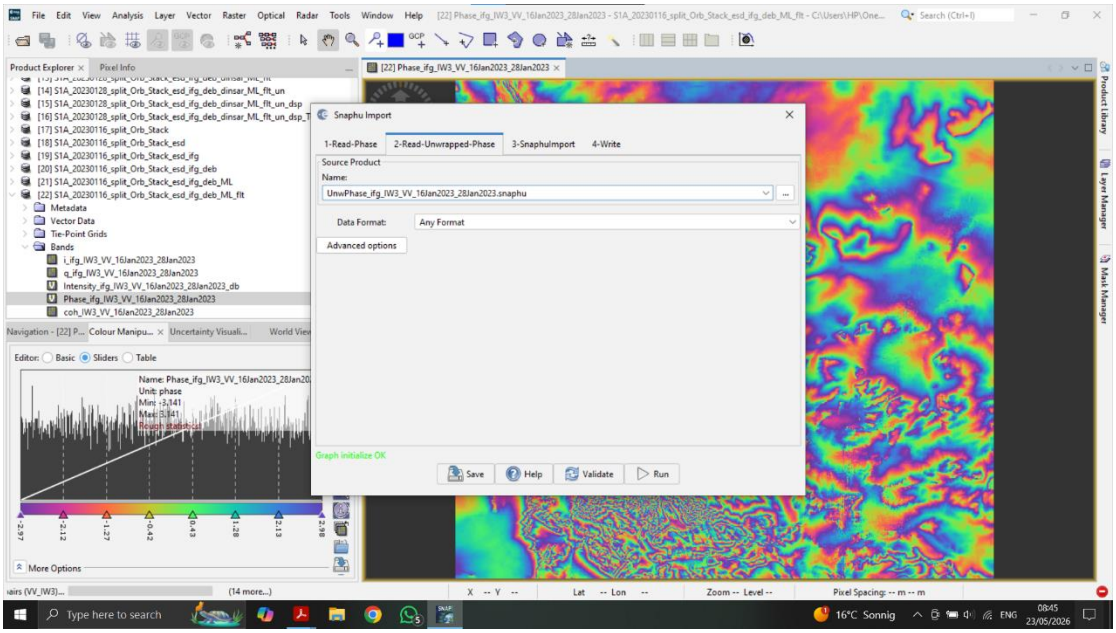


Figure 71. Selecting the corresponding file as input in "2-Read-Unwrapped-Phase" (Author)

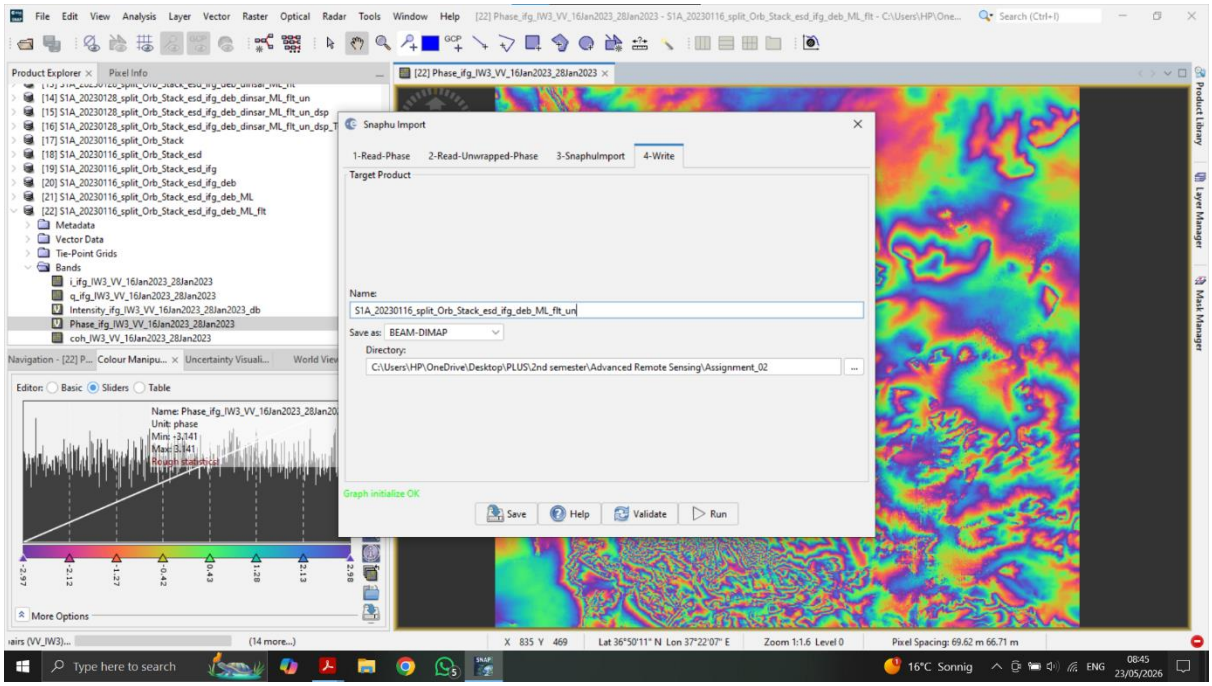


Figure 72. Adding "un" to the target product name in "4-Write" (Author)

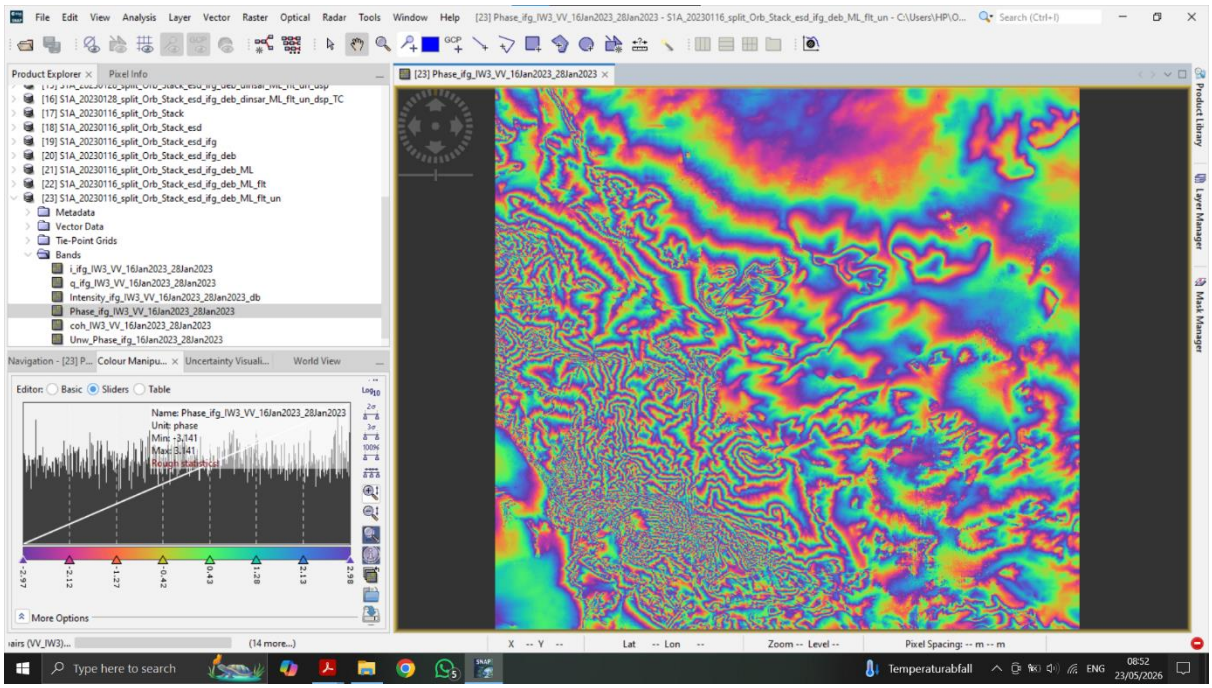


Figure 73. Representation of the unwrapped interferometric phase (Author)

I) Phase to Elevation

Next step was to convert the unwrapped phase into elevation values, that is creating DEM. It was selected “Radar”, “Interferometric”, “Products” and then “Phase to Elevation”. The unwrapped phase product was selected as the input. In the processing parameters, the DEM was changed to “SRTM 1Sec HGT (Auto Download)”.

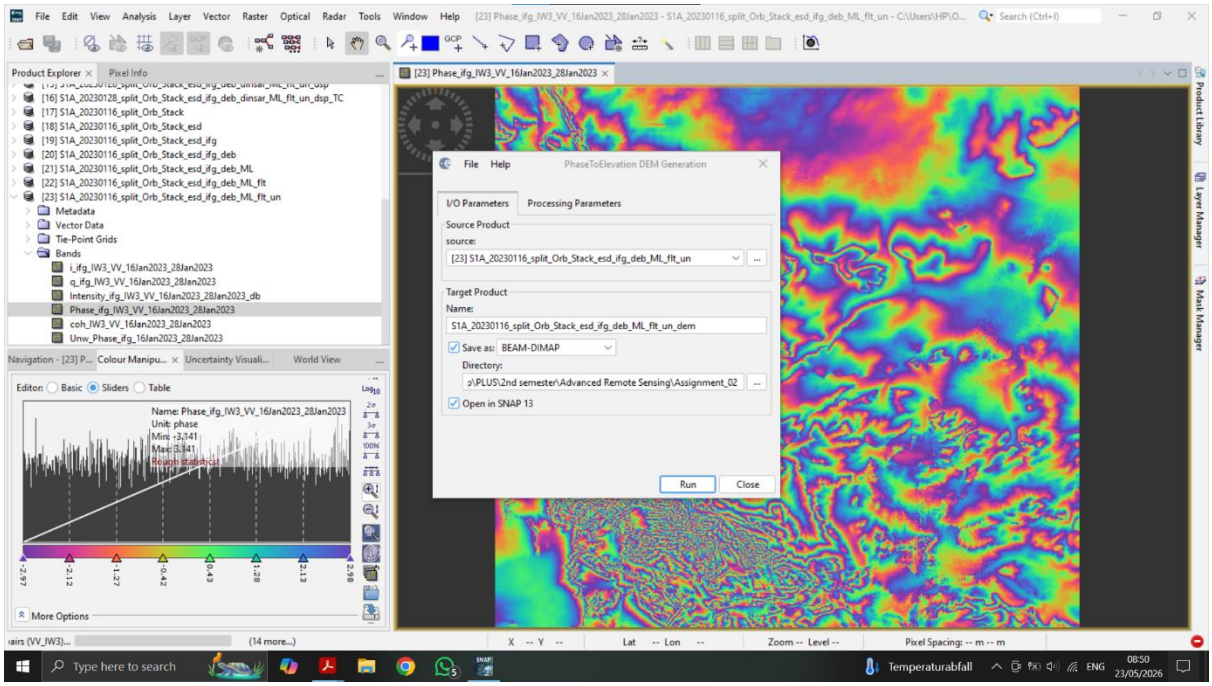


Figure 74. Selecting the unwrapped phase product as input (Author)

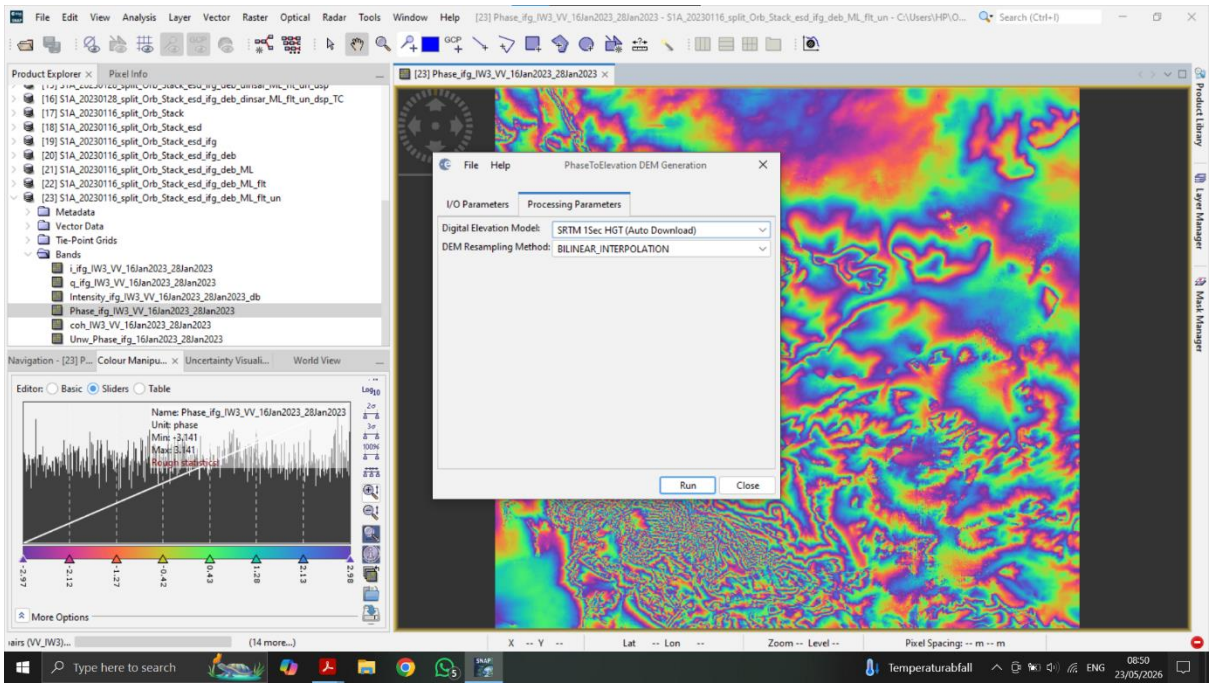


Figure 75. Changing DEM to "SRTM 1Sec HGT (Auto Download)" (Author)

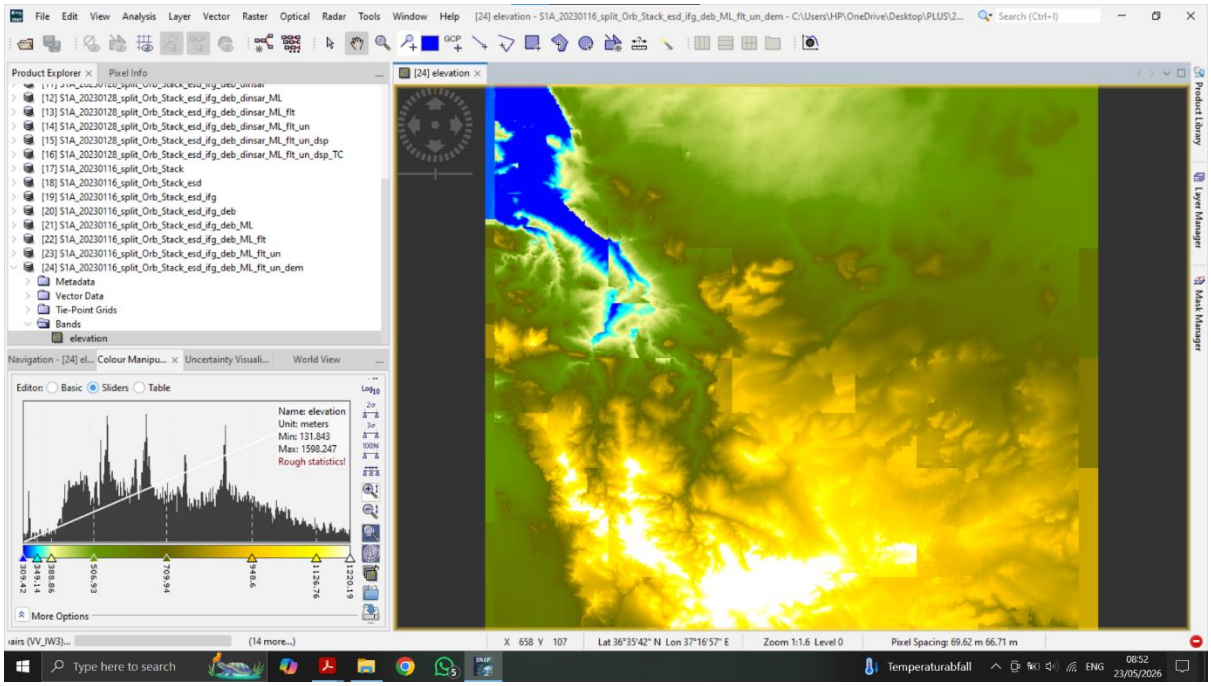


Figure 76. Representation of the newly created DEM (Author)

m) Terrain Correction

The last step was to georeference the generated DEM and remove the distortions caused by topography, to enable consistent analysis of the map. It was selected “Radar”, “Geometric”, “Terrain Correction” and then “Range-Doppler Terrain Correction”. The product from the last step was selected as input. In “Processing Parameters”, DEM was changed to “SRTM 1Sec HGT (Auto Download)”. The rest was left as it is.

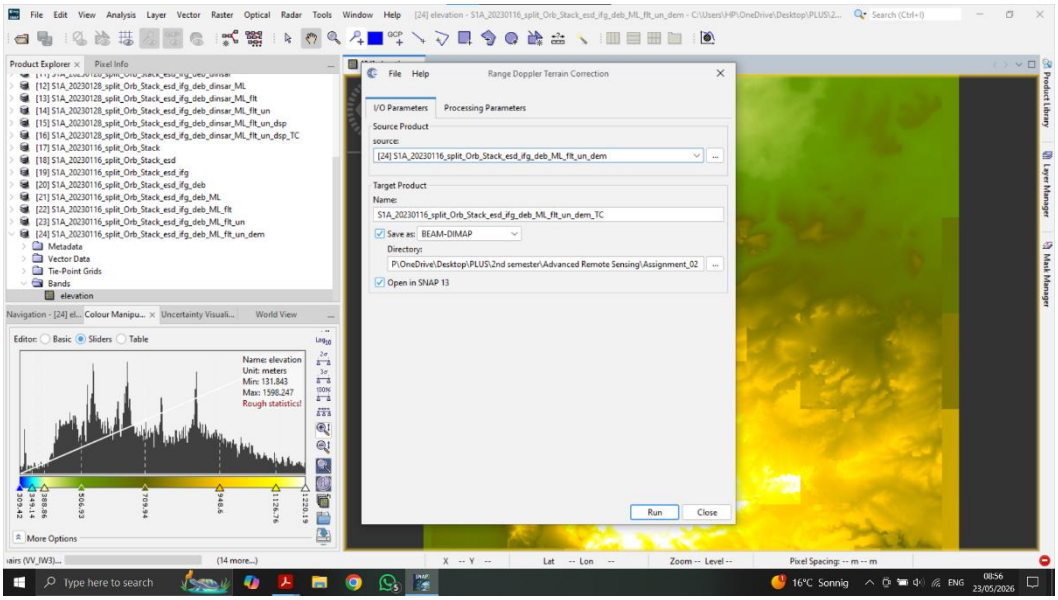


Figure 77. Selecting DEM from the last step as the input (Author)

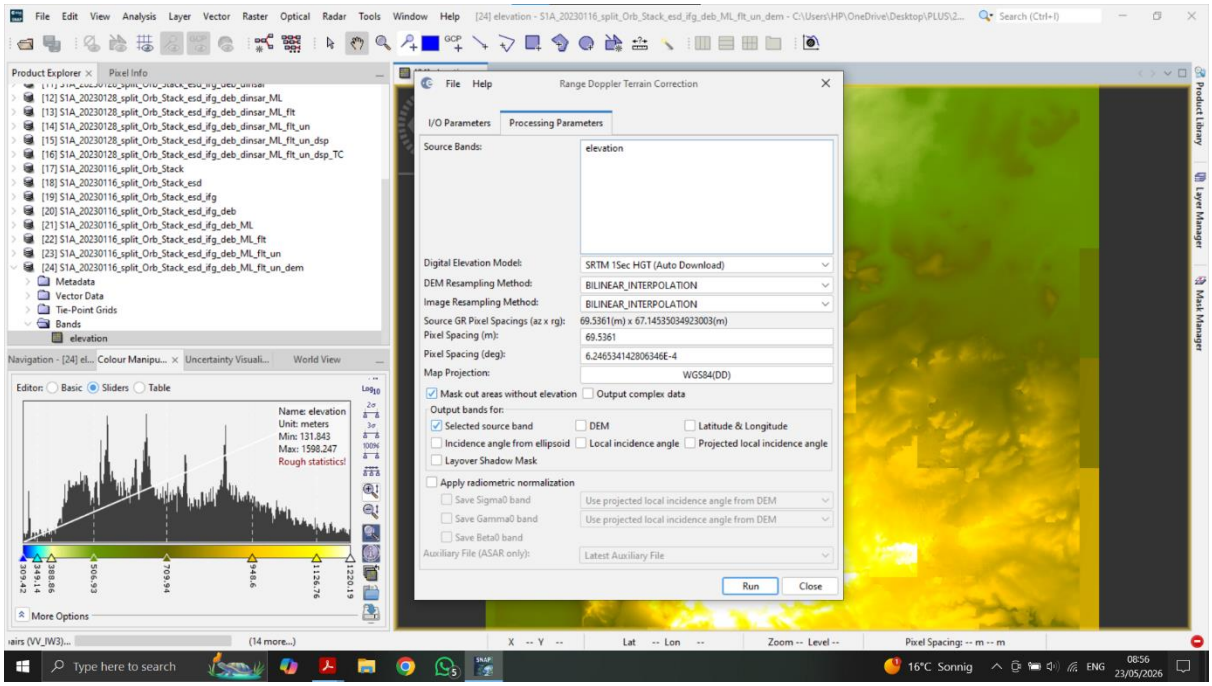


Figure 78. Changing DEM to "SRTM 1Sec HGT" (Author)

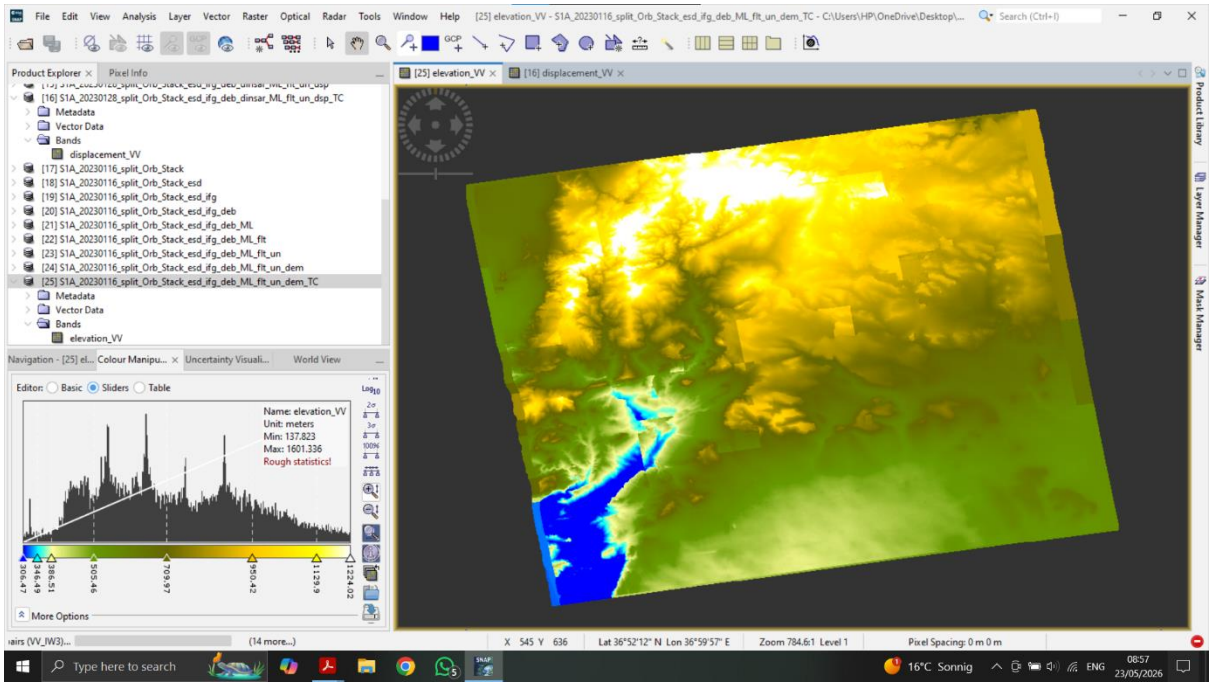


Figure 79. The resulting georeferenced and corrected DEM (Author)

As presented in DEM, the elevation values range from 137.82-1601.34 meters. The higher elevations are presented in white and light yellow colors and can be observed in upper central part of DEM. The lower elevations are presented in blue color and can be observed in lower left part of DEM. Parts where (topographic) interferometric fringes were more dense (have more elevation changes) can be observed as upper

central, upper right and lower left parts of DEM (white, bright yellow and blue colored areas). This means that terrain (elevation) change is more pronounced across smaller space (not due to surface deformation), for example, high to high, high to low or low to low. Parts where (topographic) interferometric fringes were less dense (have less elevation changes) can be observed as upper left corner and lower central to lower right parts of DEM (areas represented with a mixture of yellow and green colors).

According to the displacement map, the biggest surface displacements (by absolute value) occurred in the left upper part of AOI, which is area with approximately 500-800 meters elevation (also catches area with higher elevation over 1000 m and more dense topographic phase fringes, but in smaller portion). Displacement map presents surface deformations, while DEM presents elevation changes.

6. Credits

This assignment was completed using theoretical and practical materials provided by Professor Dr. Karima Hadj-Rabah on the e-learning platform Blackboard.

EndNote sofware was used for referencing.

7. References

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